Depth Fingerprinting: Classifying Recursive Generation Depth in Base-Model LLM Text

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Abstract

LLM-generated text is increasingly recycled into training corpora—an estimated 7–17% of recent scientific and peer-review content is now machine-produced (Liang et al., 2024b,a)—yet detection methods pose only a binary human-or-machine question, ignoring *generation depth*: how many recursive LLM steps a text has undergone. We show that successive re-generation compresses *where* text sits in distributional space while leaving *how* it is shaped nearly unchanged, an asymmetry we term *differential depth response* (DDR). Across eight base-model generators spanning five families (0.6B–8B parameters) under nucleus sampling (p=0.9, $\tau \leq 0.7$), NLL location and lexical diversity drift monotonically with depth (Cohen's d=0.76–1.17) while core distributional shape features remain near-stationary ($d \leq 0.10$); per-generator validation, convergent effect sizes, and greedy-decoding replication confirm this partition. Prior work on iterative generation documents *what converges*; the central finding here is *what is preserved*: shape invariance provides a cross-generator reference for depth estimation without per-generator calibration. A lightweight 7-feature ensemble exploiting DDR achieves leak-free binary F1 = 0.694 (gen-1 vs. gen-2+) under leave-one-generator-out evaluation, matching multi-detector ensembles at $2.4\times$ lower cost; an end-to-end classifier reaches 3-class F1 = 0.905 (6-gen LOGO), establishing the analytical framework as the primary contribution. Instruction tuning compresses DDR, bounding current applicability to base models.

1 Introduction

An estimated 7–17% of recent scientific publications already contain substantial LLM-modified content (Liang et al., 2024b,a). As these outputs re-enter training corpora (Soldaini et al., 2024; Penedo et al., 2024) and seed further generation, text accumulates at varying *generation depths*; each successive cycle moves distributional properties further from both the human source and the first-generation predecessor. State-of-the-art detectors (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Narayanasamy et al., 2026) answer only the binary question *human or machine*, discarding the depth dimension that data provenance (Wenzek et al., 2020; Xie et al., 2023) and model collapse analysis (Shumailov et al., 2024; Dohmatob et al., 2025) increasingly require.[1] We find that recursive generation produces a decomposable fingerprint: distributional *location* drifts with depth while *shape* remains stable across generators, an asymmetry sufficient for cross-generator depth prediction.

Distributional compression under recursive generation is well-documented (Guo et al., 2024; Mohamed et al., 2025; Geng et al., 2026), yet existing analyses characterize *that* and *why* convergence occurs without decomposing what drifts from what is preserved. We introduce *depth fingerprinting*, classifying text by ordinal generation depth (gen-0 / gen-1 / gen-2+), and decompose this compression into structurally distinct components: across eight base-model generators spanning five architectural families (0.6B–8B), the NLL distribution's central tendency (mean entropy, lexical diversity) drifts monotonically with depth while its shape (entropy extremes, tail structure) remains an order of magnitude more stable (d=0.76–1.17 vs. $d < 0.10$ for near-stationary features; boundary features show intermediate stability; convergent effect-size measures confirm this partition). We term this location/shape asymmetry the *differential depth response* (DDR; Figure 2), a structural property of fixed-parameter autoregressive sampling that is mechanistically distinct from the tail-coverage erosion in model collapse (Shumailov et al., 2024). The stationary shape component provides the cross-generator reference needed for depth prediction without per-generator calibration.

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[1] We scope this study to base (pretrained) models, which remain prevalent in synthetic data production and corpus construction; instruction-tuned models are analyzed in Appendix k.

As a proof of concept, a lightweight 7-feature ensemble informed by DDR achieves leak-free binary F1 = 0.694 under leave-one-generator-out evaluation, matching multi-detector ensembles at $2.4\times$ lower cost; an end-to-end QLoRA classifier reaches 3-class F1 = 0.905 (6-gen LOGO; Appendix e), confirming that the depth signal extends well beyond handcrafted features.

Our contributions:

1. **Empirical decomposition (DDR).** Across eight generators from five families, we decompose inference-chain drift into a location component that shifts monotonically (d=0.76–1.17) and a shape component that remains near-stationary ($d < 0.10$). Where prior work documents *what converges* (Guo et al., 2024; Geng et al., 2026), DDR identifies *what is preserved* and enables depth detection without per-generator calibration.
2. **Task formulation and evaluation suite.** We define generation depth classification as an ordinal prediction task and construct a controlled dataset of 37K texts from eight generators across five families (0.6B–8B), with leave-one-generator-out cross-validation as the primary evaluation protocol.
3. **Scope characterization.** Systematic evaluation across temperature, prompt type, instruction tuning, and model scale maps the operational envelope of depth fingerprinting, identifying key challenges in instruction-tuned models, cross-domain transfer, and perturbation robustness.

2 Related Work

2.1 AI-Generated Text Detection

The detection literature spans statistical tests (Gehrmann et al., 2019; Zellers et al., 2019; Solaiman et al., 2019), zero-shot scoring (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Yang et al., 2024), surprisal diversity (Basani and Chen, 2026), geometric invariants (Tulchinskii et al., 2023), and learned classifiers (Verma et al., 2024; Mady et al., 2026; Sadasivan et al., 2025; Narayanasamy et al., 2026); see Wu et al. (2025) and Jawahar et al. (2020) for surveys. RAID (Dugan et al., 2024) benchmarks binary detection across generators and attacks; we extend the task from binary to ordinal depth estimation. Several recent efforts move beyond the binary boundary (collaboration roles (Cheng et al., 2025), joint detection-attribution (Rao et al., 2025), creator-editor modes (Li et al., 2026), multi-generator attribution (Wang et al., 2024)) but all taxonomize by *source* or *production mode*, not recursive generation *depth*; repurposing their scores as depth predictors reveals high cross-generator variance (binary F1 = 0.54–0.71; Table 3) (Krishna et al., 2023; Chakraborty et al., 2024). Our contribution is orthogonal: ordinal depth classification (Gutiérrez et al., 2016; Diaz and Marathe, 2019; Cao et al., 2020; Kasa et al., 2024) as a task, and the finding that combining drifting and stationary features yields consistent cross-generator performance where individual detection scores show high variance (Uchendu et al., 2020; Munir et al., 2021).

2.2 Model Collapse and Iterative Generation

Self-consuming generative models progressively lose distributional coverage (Alemohammad et al., 2024; Shumailov et al., 2024; Briesch et al., 2023; Seddik et al., 2024), though data mixing mitigates collapse (Gerstgrasser et al., 2024; Dohmatob et al., 2025). In particular, the autophagous loop formalized by Alemohammad et al. (2024) mirrors the recursive generation setup we construct at inference time, and Dohmatob et al. (2024) provide a theoretical account showing that even minimal synthetic-data contamination alters scaling laws—predicting the distributional compression that DDR empirically decomposes into drifting and stationary components. These studies address *training-loop* degradation where parameter updates erode tail coverage; indeed, Schaeffer et al. (2025) identify eight distinct definitions of "collapse," most tied to retraining dynamics. We examine *inference chains* where parameters are fixed and only the input distribution evolves. Two concurrent works are most directly relevant: Mohamed et al. (2025) analyze paraphrasing and translation chains, documenting progressive information loss, while Geng et al. (2026) demonstrate attractor dynamics that bound distinguishable depth to approximately three classes; Wang et al. (2025) provide complementary evidence that LLM outputs converge to cyclic attractor states under repeated generation. We study *non-paraphrastic continuation* chains, where the absence of semantic anchoring permits richer depth signal, and offer a complementary decomposition: where these works characterize *that* and *why* convergence occurs, DDR isolates *what* converges (NLL location) from what is preserved (NLL shape), enabling ordinal classification without per-generator calibration. The independent convergence of all three approaches on $\sim$3 distinguishable depth classes (via Markovian dynamics (Geng et al., 2026), information-theoretic loss (Mohamed et al., 2025), and distributional feature compression) points to a shared resolution ceiling in recursive text generation (Table 45 in Appendix aa.6 provides a structured comparison).

2.3 Vocabulary Diversity and Stylometry

Declining lexical diversity in LLM text is documented in both training (Guo et al., 2024) and deployment settings (Padmakumar and He, 2024); MAUVE (Pillutla et al., 2021) provides a divergence-based metric for quantifying this distributional gap. Guo et al.'s finding that recursive training progressively impoverishes vocabulary aligns with the monotonic decline of TTR and hapax ratio we observe across generation depths; these surface features constitute the drifting component of DDR. More broadly, our surface features are classical stylometric measures (Stamatatos, 2009; Koppel et al., 2009); we repurpose them from generator identity (Uchendu et al., 2020) to the depth dimension, where their monotonic decline under top-p sampling (Holtzman et al., 2020) provides a complementary signal to reference-model statistics.

2.4 Data Quality and Provenance

Scalable data curation relies on quality signals (Xie et al., 2023; Wettig et al., 2024; Wenzek et al., 2020); generation depth offers a complementary provenance axis. Proactive watermarking (Kirchenbauer et al., 2023; Dathathri et al., 2024; Aaronson and Kirchner, 2022) embeds identifiable patterns at generation time but requires generator cooperation and degrades across multi-step chains (Ameen et al., 2026); our approach is entirely post-hoc, requiring no modification to the generation process and remaining applicable when watermarking is absent or has washed out. Our NLL-based analysis connects to Shi et al. (2024) (probability-tail membership inference) but extends from binary membership to full distributional decomposition for ordinal depth. Koike et al. (2025) formalize this connection from the opposite direction, showing that detection scores implicitly perform membership inference, underscoring the shared statistical basis of detection and provenance.

Feature	Source	Measures
entropy_mean	Ref. NLL	Central tendency of token surprise
entropy_std	Ref. NLL	Spread of token surprise
entropy_max	Ref. NLL	Maximum token surprise
entropy_min	Ref. NLL	Minimum token surprise
tail_decay	Ref. NLL	Log-prob decay slope (ranks 100–1K)
TTR	Surface	Type-token ratio
hapax_ratio	Surface	Proportion of hapax legomena

Table 1: Seven distributional features. Reference features use Pythia-410M per-token NLL; surface features require no reference model. DDR analysis (§4.2) reveals that entropy mean and the two surface features (TTR, hapax ratio) drift with depth while entropy max, min, and tail decay remain near-stationary; entropy std is intermediate.

3 Method

3.1 Task Formulation

Given a text x, we predict its *generation depth* $d \in \{0, 1, 2+\}$: human-written (d=0), first-generation LLM output (d=1), or text produced by two or more recursive LLM generations (d=2+). A generation chain of depth k is defined recursively: $x^{(0)} \sim \mathcal{C}_0$ (human corpus), $x^{(k)} = G(x^{(k-1)})$ for $k \geq 1$, where G denotes stochastic autoregressive generation. Depths beyond 2 are merged into 2+ because feature convergence compresses inter-class margins at deeper levels (§4.2).

3.2 Generation Chain Construction

We construct generation chains using eight generators spanning five model families: **Pythia** (2.8B, 1B) (Biderman et al., 2023), **OLMo** (OLMo-2-1B) (Groeneveld et al., 2024), **Qwen** (Qwen3-1.7B, Qwen3-0.6B, Qwen3-8B-Base) (Qwen Team, 2025), **Llama** (Llama-3.2-3B) (Grattafiori et al., 2024), and **Mistral** (Mistral-7B-v0.3) (Jiang et al., 2023). These generators span five distinct architectures with a 13× parameter range (0.6B–8B), trained on heterogeneous corpora, ensuring that observed depth patterns reflect properties of recursive generation rather than idiosyncrasies of any single model family or training distribution.

For each generator, we sample approximately 1,000 seed passages from English Wikipedia as $\mathcal{C}_0$. The same pool of seed passages serves as gen-0 for all generators. We use shared seeds across generators so that cross-generator comparisons control for input content and isolate the effect of the generation model from source text variation. In

LOGO evaluation, gen-0 samples from the held-out generator's chains also appear (with the same text) in the training generators' chains, creating a potential train–test overlap for the gen-0 class. We address this with a supplementary binary evaluation excluding gen-0 entirely (§4.3). Each seed is iteratively fed through the generator to produce a chain of depth 4 (i.e., gen-0 through gen-4), using the prompt "`Continue the following text: {text}`". We generate up to depth 4 to empirically confirm convergence: 5-class analysis shows adjacent depths 2–4 become indistinguishable (F1 $\approx$ 0.27; Appendix d), motivating the gen-2+ class collapse. All generation uses temperature τ=0.7, nucleus sampling with p=0.9, and maximum length of 256 new tokens. After filtering chain breaks and passages too short for feature extraction, 37,058 texts remain across all depths (Appendix a); depths 2–4 are collapsed into 2+.

3.3 Feature Extraction

The seven features operationalize two complementary views of distributional drift: five reference-model features systematically cover location, spread, extremes, and tail structure of the token-level NLL distribution relative to a fixed model, while two surface features capture lexical diversity contraction without requiring any reference model, providing a model-free cross-check on depth signal. This dual-view design enables the location/shape decomposition that constitutes our primary finding. We deliberately use interpretable distributional statistics so that the DDR partition remains transparent and mechanistically informative; end-to-end neural features improve classification (+11.6 pp, 6-gen LOGO; Appendix e) but obscure the decomposition. Cross-group ablation (§4.2) and leave-one-feature-out analysis (Appendix f) confirm that both feature groups contribute independently.

We extract seven distributional features per text (Table 1), selected to provide principled coverage of the NLL distribution—location (mean), spread (std), extremes (max, min), and tail structure (decay)—complemented by two reference-free surface diversity measures. We evaluated six additional NLL statistics (median, 90th percentile, IQR, CV, skewness, kurtosis) as candidate features; all exhibit $|r| > 0.65$ with existing features and yield no classification gain when added (ΔF1 ≤ -0.011; Appendix f.1).

Reference-model features. A held-out reference model (Pythia-410M, chosen to be smaller than all generators) computes per-token negative log-likelihood (NLL) for each text. We use "entropy" as shorthand for token-level NLL throughout, following convention in prior detection work; these are point-wise NLL values, not Shannon entropy of the full distribution. Although Pythia-410M shares the GPT-NeoX architecture with two of the eight generators, cross-family reference model experiments (§4.2) confirm that depth features are not reference-specific (LOGO F1 varies by $\leq$0.013 across Pythia, OLMo, and Qwen references). From the resulting NLL sequence, we extract five statistics: mean, standard deviation, maximum, and minimum of the token-level NLL, plus tail_decay—the slope of the vocabulary-level averaged log-probability distribution in the mid-rank region (ranks 100–1000), capturing how steeply probability mass decays beyond high-frequency tokens.

Surface features. Two features capture lexical diversity without requiring a reference model: type-token ratio (TTR) and the proportion of words appearing exactly once (hapax_ratio).

Feature complementarity. These features exhibit a *differential depth response* (DDR), separating into drifting and non-drifting groups (§4.2; pairwise correlations in Appendix s). TTR and hapax ratio are strongly correlated ($r = 0.97$); both are retained for marginal LOFO gains.

3.4 Evaluation Protocol

Cross-validation. We evaluate generalization via **Leave-One-Generator-Out (LOGO)** cross-validation: each fold holds out one generator entirely, so the classifier must generalize to a generator absent from training—the strictest per-model protocol for assessing deployment against previously unseen generators. LOGO does not isolate family-level effects; **Leave-One-Family-Out (LOFO)** addresses this by holding out entire model families (binary F1 = 0.686; §4.1). Generator subsets vary across experiments (Appendix x).

Classifier. We use XGBoost (Chen and Guestrin, 2016) throughout (n_estimators=200, max_depth=6, learning rate 0.1; fixed without tuning), with identical hyperparameters across all methods in Table 3. Alternative classifiers (CORAL (Cao et al., 2020), CORN (Shi et al., 2023), Frank-Hall (Frank and Hall, 2001)) yield comparable performance (Appendix h); discriminative power resides in the features, not the classifier.

Metrics and statistical testing. The primary metric is **macro-averaged F1** over the three depth classes, with 95% bootstrap CIs (n=1,000). All p-values are two-sided with Bonferroni correction within each family of simultaneous tests.

Baselines. We compare the 7-feature ensemble against four single-score detection methods repurposed as depth predictors: Avg-LogProb, FastDetectGPT (Bao et al., 2024), Binoculars (Hans et al., 2024), and LD-Score (Narayanasamy et al., 2026). Each baseline score is used as a univariate feature in the same XGBoost framework under LOGO cross-validation to ensure a controlled comparison.

Length confound control. Since TTR and hapax_ratio are mechanically length-dependent, we verify depth signal is not a length artifact via length-matched subsampling that equalizes length distributions across depth classes (§4.1; Appendix b).

4 Results

4.1 Depth Classification Performance

The 7-feature ensemble achieves 3-class LOGO F1 = 0.762 (95% CI [0.758, 0.766]; N = 37,058) across eight generators from five families (Table 2), with cross-generator F1 spread of 9.1 pp capturing generator selection uncertainty. Because gen-0 seeds are shared across generators (§3.2), gen-0 classification is near-perfect (per-class F1 ≈ 0.94), inflating the 3-class metric. We therefore adopt **leak-free binary F1** (gen-1 vs. gen-2+) = 0.694 (CI [0.688, 0.700]) as the primary metric throughout: it excludes gen-0 entirely and is unaffected by seed sharing. Per-generator binary F1 ranges from 0.633 (Mistral-7B) to 0.724 (Pythia-2.8B; SD = 0.028), with the primary difficulty concentrated at the gen-1 / gen-2+ boundary (Appendix g). CIs reflect bootstrap sampling variability; the cross-generator range (9.1 pp) captures the distinct and larger uncertainty of generator selection.

Cross-model generalization. Cross-model chains (randomly selected generator per step) yield binary F1 = 0.830 (95% CI [.818, .841]; per-class breakdown in Appendix o) under 5-generator LOGO.[2] Single-generator transfer (Appendix z) establishes that the signal is not a mixed-chain artifact: classifiers trained on one generator transfer to all others with mean binary F1 = 0.810; LOFO yields 0.686. A scale effect emerges: Pythia achieves the highest F1 (0.718–0.724) while larger models degrade (Mistral-7B: 0.633), partly reflecting same-family affinity with the Pythia-410M reference (cross-reference spread ≤0.013; §4.2); cross-domain signal persists with variation (Appendix j). An end-to-end QLoRA classifier achieves 3-class F1 = 0.905 under 6-generator LOGO (+11.6 pp vs. the same-subset XGBoost baseline of 0.789; Appendix e), confirming that richer token-level representations capture additional depth signal, albeit requiring per-generator fine-tuning (DDR's handcrafted features are generator-agnostic); on instruct data, QLoRA recovers to 0.89–0.91 within-model (Appendix k), establishing the decomposition, not the classifier, as DDR's primary contribution.

Length confound control. Length-matched subsampling yields binary F1 = 0.724 (vs. 0.770 unmatched); a random-subsample control verifies the drop reflects deconfounding, not sample-size reduction (Appendix b).

Seed-sharing control. Seed-level demeaning (F1 = 0.701) and an independent-seed experiment with disjoint seed pools (F1 = 0.680; Appendix u) both verify that the classifier captures generative degradation, not seed-specific content.

4.2 Differential Depth Response

Formal definition. For feature f measured across depths $d \in \{0, 1, 2+\}$, we define its *depth sensitivity* as $\text{DDR}(f) = |\rho(\text{depth}, f)|$, where ρ is Spearman correlation over all per-text measurements. A feature is classified as *drifting* if Cohen's $d > 0.50$ between adjacent depths, or *stationary* if $d < 0.15$; these thresholds serve as reference anchors from behavioral science, with the partition independently validated by data-driven k-means clustering (silhouette = 0.660, p=0.030 vs. null; Appendix w). Under autoregressive generation with nucleus sampling, the top-p truncation preserves the conditional distribution's functional form while inputs grow increasingly self-typical—we hypoth-

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[2] 5-generator subset (Pythia-2.8B/1B, OLMo-2-1B, Qwen3-0.6B/1.7B); generator-level LOGO with bootstrap CIs (n = 2,000). Not directly comparable to 8-generator LOGO due to differing subsets.

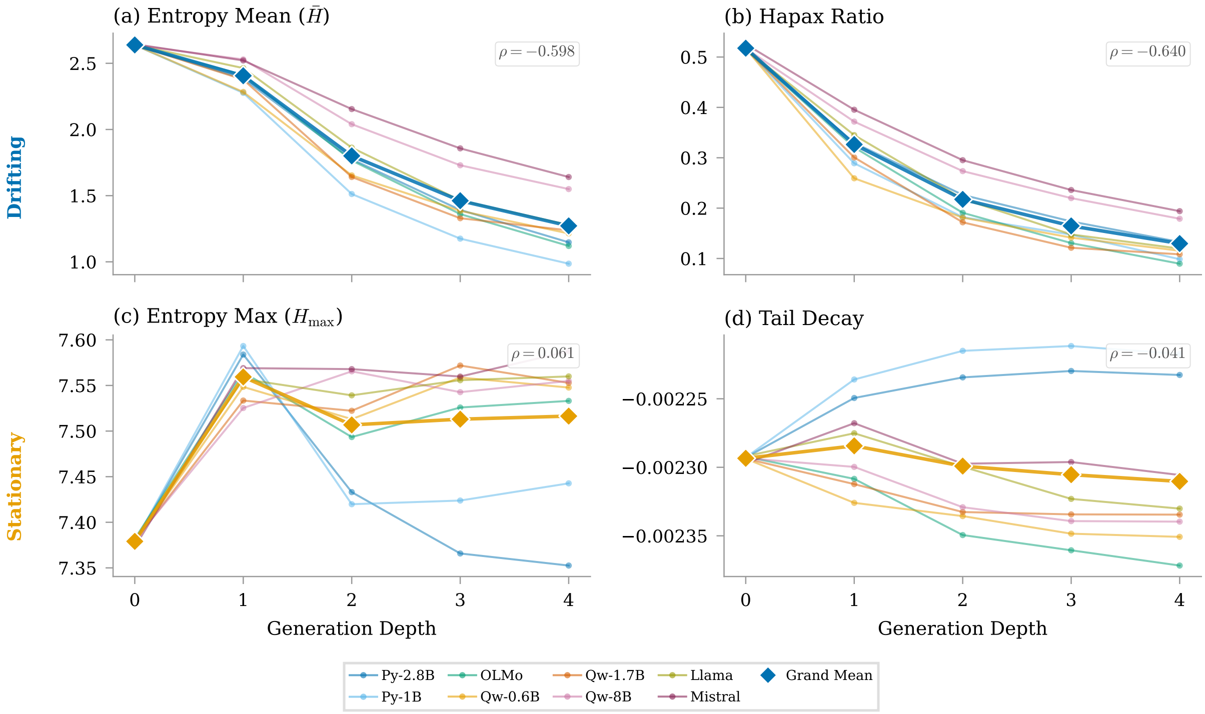

Figure 1: Feature trajectories across generation depth. **Top row (drifting)**: entropy mean and hapax ratio decline monotonically across all eight generators ($|\rho| > 0.59$). **Bottom row (stationary)**: entropy max and tail decay remain flat ($|\rho| < 0.07$). Colored lines: individual generators; diamonds: cross-generator mean (blue for drifting, amber for stationary). This dichotomy defines the differential depth response (DDR).

Model	**Family**	**Params**	**Bin F1**	**95% CI**	**3-class F1**†
Pythia-2.8B	Pythia	2.8B	**0.724**	[.707, .742]	0.798
Pythia-1B	Pythia	1.0B	0.718	[.700, .735]	**0.806**
Qwen3-1.7B	Qwen	1.7B	0.709	[.693, .726]	0.772
Llama-3.2-3B	Llama	3.2B	0.697	[.680, .713]	0.775
OLMo-2-1B	OLMo	1.5B	0.695	[.679, .711]	0.781
Qwen3-8B	Qwen	8.2B	0.692	[.674, .708]	0.714
Qwen3-0.6B	Qwen	0.6B	0.691	[.673, .709]	0.780
Mistral-7B	Mistral	7.0B	0.633	[.617, .648]	0.682
Average (8-gen LOGO)			**0.694**	[.688, .700]	**0.762**
LOFO (family-level)			0.686	—	—

Table 2: 8-generator LOGO macro-F1 for the 7-feature ensemble (bootstrap 95% CIs, $n = 1000$). Binary F1 (gen-1 vs. gen-2+) is the primary metric; it excludes gen-0 entirely and is unaffected by seed sharing. Cross-generator binary F1: SD = 0.028, IQR = [0.692, 0.714]. †3-class F1 carries upward bias because gen-0 seeds are shared across generators (§3.2); independent-seed controls confirm the depth signal (F1 = 0.680; Appendix u).

esize this reflects the mixing property of finite-vocabulary autoregressive processes (Geng et al., 2026)—predicting that location statistics should drift while shape statistics remain invariant.

The classification results above establish *that* depth is distinguishable; we now ask *why* by examining which features respond to recursive generation. Figure 1 reveals a consistent dichotomy across all eight generators: central-tendency features (entropy mean, TTR, hapax ratio) decline monotonically with depth, while shape features (entropy max, tail decay, entropy min) remain flat. Appendix y illustrates the DDR with a concrete generation chain. The primary partition is binary: k=2 clustering cleanly separates three drifting from four non-drifting features; higher k degrades quality (k=3: silhouette = 0.532; Appendix w). A domain-informed refinement resolves entropy std as a sole *intermediate* member (d=0.55, $|\rho|$=0.16), distinct from the three near-stationary features ($d < 0.10$; CV shift <3% vs. >15% for drifting features). Multiple effect-size measures corroborate this partition (Quantification below).

Pooled Spearman correlations with depth corroborate the binary split (Figure 2): drifting features show substantial monotonic trends (hapax ratio $\rho = -0.64$, TTR $\rho = -0.63$, entropy mean $\rho = -0.60$), while non-drifting features remain near-zero (tail decay $|\rho| = 0.04$; per-depth distributions in Appendix p). The six-fold magnitude gap between drifting ($|\rho| > 0.60$) and non-drifting features ($|\rho| \leq 0.16$) defines the DDR partition. Within the non-drifting group, entropy std ($|\rho| = 0.16$, d=0.55) shows a substantial level shift despite low monotonicity, distinguishing it

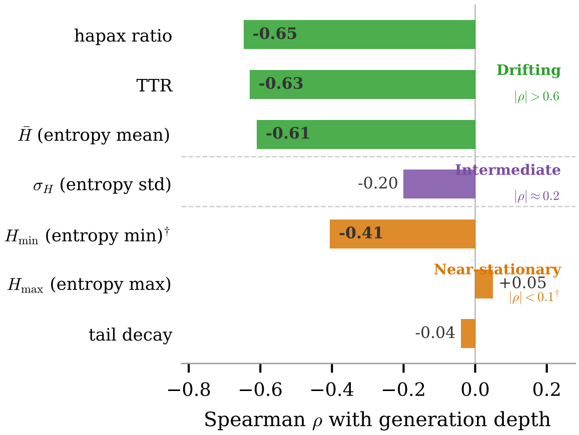

Figure 2: Differential depth response (DDR). Diamonds: cross-generator means; dots: per-generator values. Drifting features (blue; $|\rho| > 0.60$, d=0.76–1.17) show strong monotonic depth sensitivity; non-drifting features remain stable. Entropy std (wine; d=0.55) is an intermediate member; near-stationary features (amber; $d \leq 0.10$) show negligible drift. Entropy min ($|\rho|$=0.40 pooled; gen-0/gen-1 sensitive) and tail decay (TOST 4/8) are boundary features within this group.

from the three near-stationary features ($d < 0.10$); this refinement is stable across d thresholds in $[0.56, 0.75]$ (Appendix w).

Quantification. Parametric and non-parametric effect-size measures converge on the split. Cohen's d (Cohen, 1988) shows a $\geq$7× gap between drifting (d=0.76–1.17) and near-stationary features ($d \leq 0.08$); KS statistics, Glass's Δ, Cliff's δ, and CV shift all corroborate (Appendix w), with the partition robust under heteroscedasticity (Glass's Δ gap $\geq$9.5×; Appendix w). Four features fall cleanly into drifting or non-drifting groups. Entropy min and tail decay are *boundary features*: entropy min is near-stationary for gen-1 vs. gen-2+ (d=0.016) but sensitive at gen-0/gen-1 ($|\rho|$=0.40); tail decay is *weakly stationary*, passing TOST for only 4/8 generators ($|d|$=0.10–0.22). Entropy std (d=0.55, $|\rho|$=0.16) is the sole intermediate member.

Validation. Per-generator split analysis provides the strongest validation: each of the eight holdout generators independently recovers the identical drifting/non-drifting partition (per-generator silhouette range 0.505–0.760, invariant across $\varepsilon \in [0.10, 0.75]$; Appendix w). Per-generator TOST equivalence testing (ε=$|d|$<0.20; gen-1 vs. gen-2+; power is limited with only three depth levels) confirms near-stationarity for entropy min (8/8 pass) and entropy max (6/8); tail decay passes for only 4/8, with $|d|$=0.10–0.22 spanning the boundary. A permutation test (n=10,000) establishes the split as non-random (null mean = 0.46); exhaustive enumeration of all $\binom{7}{3}$=35 possible splits ranks this partition first (Appendix w), and disjoint holdout splits recover identical zone assignments (Appendix w). Reclassifying either boundary feature as intermediate preserves the core drifting/non-drifting split and leaves classification unchanged.

Feature complementarity. The full 7-feature set (F1 = 0.773, 10-fold stratified CV, complementing the LOGO protocol in Table 2) outperforms surface-only (0.688) and reference-only (0.713; McNemar $p \approx 0$; Appendix r). PCA confirms the three drifting features share a single latent dimension (92.3% variance; Appendix s); replacing them with PC1 costs only 1.1 pp, reinforcing that DDR's contribution is identifying *which* distributional properties drift, not classification per se. TTR and hapax ratio ($r = 0.97$) are both retained for marginal LOFO gains despite feature masking (Strobl et al., 2008) (Appendix f).

Cross-family reference models. To isolate reference-model bias, we evaluate on a 5-generator subset excluding all Pythia-family generators from the test set—a stricter protocol than full 8-gen LOGO, which includes same-family generators that could inflate F1. Alternative references (OLMo-2-1B, Qwen3-0.6B) achieve comparable performance (F1 spread = 0.013; Table 26 in Appendix t), with drift direction consistently negative across all three references. Residual same-family proximity may inflate Pythia-family F1 by up to 1.3 pp, but the DDR partition—which features drift vs. which are stable—is fully preserved regardless of reference choice.

Temperature and prompt-type sensitivity. All four tested generators degrade at t=0.9 ($-$9.3 to $-$26.0 pp binary F1; Appendix h); continuation chains show the strongest drift (F1 = 0.781, LOPTO = 0.753; Appendix c). The DDR partition hierarchy is preserved across temperatures despite absolute effect-size compression (Appendix aa.1).

4.3 Comparison with Detection Baselines

Having identified which features carry depth signal, we ask whether existing detection methods—designed for binary real-vs-synthetic classification—incidentally capture the same struc-

Method	Feat.	3-class F1	Bin F1	ms/text
4-det. + surface	6	**0.795**	**0.708**	22.8
4-detector ens.	4	0.785	0.700	22.8
7-feature (ours)	7	0.774	0.705	**9.7**
Avg-LogProb	1	0.707	0.673	—
FastDetectGPT	1	0.660	0.569	—
LD-Score	1	0.630	0.675	—
Binoculars	1	0.411	0.539	—

Table 3: Depth classification (8-gen LOGO, pooled evaluation; pooled F1 is slightly higher than per-generator macro F1 in Table 2 because larger test folds from high-performing generators contribute proportionally more). Identical XGBoost configuration. 7-feature: one 410M model (905 MB); 4-det ensembles: two models, 1.4B params (2,868 MB). Individual detection scores (N/A) share the reference-model forward pass; per-method timing is not separately profiled.

ture. We repurpose four detection scores as ordinal depth predictors under identical XGBoost and LOGO conditions (Table 3). FastDetectGPT and Binoculars target binary real/fake detection, not ordinal depth estimation; Avg-LogProb provides the most meaningful single-feature baseline.

The 7-feature ensemble surpasses all individual baselines in 3-class F1 ($\Delta = 6.7$–36.3 pp; Figure 7 in Appendix m) and the pure 4-detector ensemble in binary F1 (0.705 vs. 0.700). Adding DDR's surface features to the four detectors yields the highest binary F1 (0.708), marginally above the 7-feature set at 2.4× the inference cost. This result underscores that DDR's primary contribution is not classification performance (multi-detector ensembles match or marginally exceed it) but the decomposition of depth signal into location-sensitive and shape-stable components, a structural distinction absent from both binary detection methods and model collapse analyses (§5).

5 Discussion

The differential depth response arises from a structural property of nucleus sampling: top-p truncation preserves the conditional distribution's functional form while recursive inputs grow increasingly self-typical, compressing expected NLL monotonically (Appendix aa.7; convergence trajectories in Appendix aa.2). This mechanism yields two testable predictions—increasing randomness should weaken location drift, and semantic constraints should attenuate it—both of which are confirmed: at τ=1.0, depth–NLL correlation collapses to ρ=-0.002 (Appendix h), and elaboration prompts reduce F1 to 0.730 (Appendix c), with drifting features contributing $5.5\times$ more to classification and 6/7 showing significant Granger causality (Appendix aa.3). Crucially, shape stability holds even under conditions that suppress drift: at τ=1.0 ($d < 0.15$; Appendix aa.1) and under greedy decoding across all eight generators (mean $|d|$=0.05); top-k (k=50) compresses drifting effect sizes without introducing new shape instability, with boundary violations in Pythia/OLMo traceable to model-intrinsic baselines (Appendix aa.4).

Depth constitutes a new operational dimension for corpus management that binary human-vs-machine classifiers cannot access. As a proof-of-concept, our classifier achieves gen-1 per-class F1 = 0.63, sufficient for batch contamination estimation (±3–4% at n=1,000; Appendix l) rather than per-document filtering. Beyond detection, depth-stratified curation enables provenance tracking invisible to binary classifiers and can serve as a prophylactic against model collapse (Drayson et al., 2025), though downstream training improvements remain statistically equivalent to unfiltered baselines (TOST equivalence at ε=$\pm 2\%$; Appendix n).

DDR is mechanistically distinct from model collapse (Shumailov et al., 2024; Dohmatob et al., 2025): retraining erodes tail coverage by updating weights, whereas inference-chain drift at fixed parameters shifts only location (Appendix aa). Prior work documents *that* iterative generation converges (Geng et al., 2026; Guo et al., 2024); DDR decomposes *what* drifts from what is preserved, enabling detection without per-generator calibration—a partition that extends to Gemma-2 with only -1.9 pp degradation (Appendix aa.5). The approach has clear resolution limits: three depth classes define the practical ceiling (5-class F1 ≈ 0.50; Appendix d), though QLoRA recovers substantially more on the 3-class task (F1 = 0.905; Appendix e). Instruction-tuned models compress DDR (F1 degrades 3.7–19.9 pp; Appendix k), yet QLoRA recovers performance to 0.89–0.91. Cross-domain evaluation yields binary F1 = 0.74–0.75 with degraded 3-class performance (F1 = 0.62–0.65; Appendix j), and the depth signal survives moderate human perturbation (binary F1 = 0.614 at 10% word replacement; Appendix i).

6 Conclusion

Recursive generation in base models imprints a differential depth response: central-tendency features drift with depth while shape features remain stable, enabling cross-generator depth classification (leak-free binary F1 = 0.694; end-to-end 3-class: 0.905) across eight generators from five families, replicated under deterministic decoding. The decomposition establishes depth as an operational axis for data curation and collapse prevention, complementary to model-collapse analyses that target retraining-induced degradation. More broadly, DDR suggests that fixed-parameter generation chains carry an inherent structural fingerprint that persists across architectures and scales, offering a provenance signal inaccessible to classifiers that treat all machine-generated text as a single category. Extending DDR to instruction-tuned models and developing provenance-aware data pipelines remain key challenges for translating depth fingerprinting into operational practice.

Limitations

All eight generators are base (pretrained) models; instruction-tuned variants show substantially degraded depth signal (Appendix k), and extending the method to RLHF-aligned models (which dominate real-world text) requires addressing confounds from length regulation and safety-driven distribution shaping. Our largest generator is 8.2B parameters; frontier models (70B+) remain untested and their higher fluency may further compress feature margins.

The evaluation assumes a fixed decoding regime (temperature 0.7, nucleus p=0.9); cross-temperature transfer degrades when test-time temperature exceeds training-time settings (Appendix q). All seed texts are English Wikipedia excerpts, and the reference model (Pythia-410M) has English-centric pretraining, limiting non-English applicability.

Depth resolution is bounded at three classes (gen-0 / gen-1 / gen-2+); finer-grained classification (5-class) yields F1 $\approx$ 0.50 with depths 2–4 mutually indistinguishable (Appendix d). Our cross-model chain and simulated editing experiments (§5) represent controlled laboratory settings; the gap to real-world post-processing pipelines remains untested. Top-k sampling preserves DDR for non-Pythia/OLMo generators but shows model-intrinsic boundary violations in Pythia and OLMo families; broader decoding-strategy coverage (e.g., beam search) remains future work.

Ethics Statement

All texts analyzed in this work are LLM-generated from publicly available Wikipedia seeds; no personal or sensitive data is involved. Our method estimates the generation depth of a text passage (how many recursive generation steps it has undergone), not the identity of its author. Individual authorship attribution is outside both the scope and the capability of this approach; privacy implications are minimal. We acknowledge that depth estimation tools could in principle be used to selectively suppress machine-generated text in contexts where such content is legitimate (e.g., accessibility tools, assistive writing). We encourage deployment in data governance pipelines where provenance transparency, rather than content removal, is the objective. All code, data, and trained models will be released upon publication.

Licenses. All generator models are publicly available under permissive licenses (Apache 2.0 for Pythia, OLMo, and Mistral; Qwen License for Qwen3; Llama Community License for Llama 3.2). Wikipedia seed passages are distributed under CC BY-SA 3.0.

Compute budget. All experiments were conducted on 4× NVIDIA RTX 5090 GPUs (32 GB each), totaling approximately 62 GPU-hours.

Software. Experiments used Python 3.12.3, PyTorch 2.10.0 (CUDA 12.8), Transformers 4.57.6, XGBoost 3.2.0, scikit-learn 1.8.0, NumPy 2.2.6, pandas 3.0.2, and SciPy 1.17.1.

References

Scott Aaronson and Jan Hendrik Kirchner. 2022. Watermarking GPT Outputs. Blog post.

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. 2024. Self-Consuming Generative Models Go MAD. In *Proceedings of the International Conference on Learning Representations*.

Mohd Ruhul Ameen, Akif Islam, Nadim Mahmud, and Md. Ekramul Hamid. 2026. Chainwash: Multi-Step Rewriting Attacks on Diffusion Language Model Watermarks. *arXiv preprint arXiv:2605.05503*.

Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. 2024. Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated

Text via Conditional Probability Curvature. In *Proceedings of the International Conference on Learning Representations*.

Advik Raj Basani and Pin-Yu Chen. 2026. Diversity Boosts AI-Generated Text Detection. In *Proceedings of the International Conference on Learning Representations*.

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In *Proceedings of the International Conference on Machine Learning*.

Martin Briesch, Dominik Sobania, and Franz Rothlauf. 2023. Large Language Models Suffer From Their Own Output: An Analysis of the Self-Consuming Training Loop. *arXiv preprint arXiv:2311.16822*.

Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. 2020. Rank Consistent Ordinal Regression for Neural Networks with Application to Age Estimation. *Pattern Recognition Letters*, 140:325–331.

Souradip Chakraborty, Amrit Singh Bedi, Sicheng Zhu, Bang An, Dinesh Manocha, and Furong Huang. 2024. Position: On the Possibilities of AI-Generated Text Detection. In *Proceedings of the International Conference on Machine Learning*.

Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Zihao Cheng, Li Zhou, Feng Jiang, Benyou Wang, and Haizhou Li. 2025. Beyond Binary: Towards Fine-Grained LLM-Generated Text Detection via Role Recognition and Involvement Measurement. In *Proceedings of the ACM Web Conference*.

Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences*, 2nd edition. Lawrence Erlbaum Associates.

Sumanth Dathathri, Abigail See, Sumedh Ghaisas, Po-Sen Huang, Rob McAdam, Johannes Welbl, Vandana Bachani, Alex Kaskasoli, Robert Stanforth, Tatiana Matejovicova, Jamie Hayes, Nidhi Vyas, Majd Al Merey, Jonah Brown-Cohen, Rudy Bunel, Borja Balle, Taylan Cemgil, Zahra Ahmed, Kitty Stacpoole, and 5 others. 2024. Scalable Watermarking for Identifying Large Language Model Outputs. *Nature*, 634:818–823.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs. In *Advances in Neural Information Processing Systems*.

Raul Diaz and Amit Marathe. 2019. Soft Labels for Ordinal Regression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Elvis Dohmatob, Yunzhen Feng, Arjun Subramonian, and Julia Kempe. 2025. Strong Model Collapse. In *Proceedings of the International Conference on Learning Representations*.

Elvis Dohmatob, Yunzhen Feng, Pu Yang, François Charton, and Julia Kempe. 2024. A Tale of Tails: Model Collapse as a Change of Scaling Laws. In *Proceedings of the International Conference on Machine Learning*.

George Drayson, Emine Yilmaz, and Vasileios Lampos. 2025. Machine-Generated Text Detection Prevents Language Model Collapse. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.

Liam Dugan, Alyssa Hwang, Filip Trhlík, Andrew Zhu, Josh Magnus Ludan, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. 2024. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12463–12492, Bangkok, Thailand. Association for Computational Linguistics.

Eibe Frank and Mark Hall. 2001. A Simple Approach to Ordinal Classification. In *Proceedings of the European Conference on Machine Learning*.

Sebastian Gehrmann, Hendrik Strobelt, and Alexander M. Rush. 2019. GLTR: Statistical Detection and Visualization of Generated Text. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics: System Demonstrations*.

Mingmeng Geng, Amr Mohamed, Guokan Shang, Michalis Vazirgiannis, and Thierry Poibeau. 2026. Markovian Generation Chains in Large Language Models. *arXiv preprint arXiv:2603.11228*.

Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Dhruv Pai, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David L. Donoho, and Sanmi Koyejo. 2024. Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data. In *Proceedings of the First Conference on Language Modeling (COLM)*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, and 1 others. 2024. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.

Dirk Groeneveld, Iz Beltagy, Evan Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, Shane Arora, David Atkinson, Russell Authur, Khyathi Raghavi Chandu, Arman Cohan, Jennifer Dumas, Yanai Elazar, Yuling Gu, Jack Hessel, and 24 others. 2024. OLMo: Accelerating the Science of Language Models. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Yanzhu Guo, Guokan Shang, Michalis Vazirgiannis, and Chloé Clavel. 2024. The Curious Decline of Linguistic Diversity: Training Language Models on Synthetic Text. In *Findings of the Association for Computational Linguistics: NAACL*.

Pedro Antonio Gutiérrez, María Pérez-Ortiz, Javier Sánchez-Monedero, Francisco Fernández-Navarro, and César Hervás-Martínez. 2016. Ordinal Regression Methods: Survey and Experimental Study. *IEEE Transactions on Knowledge and Data Engineering*, 28(1):127–146.

Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2024. Spotting LLMs With Binoculars: Zero-Shot Detection of Machine-Generated Text. In *Proceedings of the International Conference on Machine Learning*.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The Curious Case of Neural Text Degeneration. In *Proceedings of the International Conference on Learning Representations*.

Ganesh Jawahar, Muhammad Abdul-Mageed, and Laks V.S. Lakshmanan. 2020. Automatic Detection of Machine Generated Text: A Critical Survey. In *Proceedings of the International Conference on Computational Linguistics*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, and 1 others. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.

Siva Rajesh Kasa, Aniket Goel, Karan Gupta, Sumegh Roychowdhury, Pattisapu Priyatam, Anish Bhanushali, and Prasanna Srinivasa Murthy. 2024. Exploring Ordinality in Text Classification: A Comparative Study of Explicit and Implicit Techniques. In *Findings of the Association for Computational Linguistics: ACL*.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A Watermark for Large Language Models. In *Proceedings of the International Conference on Machine Learning*.

Ryuto Koike, Liam Dugan, Masahiro Kaneko, Chris Callison-Burch, and Naoaki Okazaki. 2025. Machine Text Detectors are Membership Inference Attacks. *arXiv preprint arXiv:2510.19492*.

Moshe Koppel, Jonathan Schler, and Shlomo Argamon. 2009. Computational Methods in Authorship Attribution. *Journal of the American Society for Information Science and Technology*, 60(1):9–26.

Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. 2023. Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval is an Effective Defense. In *Advances in Neural Information Processing Systems*.

Daniël Lakens. 2017. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, 8(4):355–362.

Yang Li, Qiang Sheng, Zhengjia Wang, Yehan Yang, Danding Wang, and Juan Cao. 2026. Beyond the Final Actor: Modeling the Dual Roles of Creator and Editor for Fine-Grained LLM-Generated Text Detection. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Weixin Liang, Zachary Izzo, Yaohui Zhang, Haley Lepp, Hancheng Cao, Xuandong Zhao, Lingjiao Chen, Haotian Ye, Sheng Liu, Zhi Huang, Daniel A. McFarland, and James Y. Zou. 2024a. Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews. In *Proceedings of the International Conference on Machine Learning*.

Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher Potts, Christopher D. Manning, and James Y. Zou. 2024b. Mapping the Increasing Use of LLMs in Scientific Papers. In *Proceedings of the Conference on Language Modeling*.

Mohamed Mady, Johannes Reschke, and Björn Schuller. 2026. Feature-Augmented Transformers for Robust AI-Text Detection Across Domains and Generators. *arXiv preprint arXiv:2605.03969*.

Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. 2023. DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature. In *Proceedings of the International Conference on Machine Learning*.

Amr Mohamed, Mingmeng Geng, Michalis Vazirgiannis, and Guokan Shang. 2025. LLM as a Broken Telephone: Iterative Generation Distorts Information. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Shaoor Munir, Brishna Batool, Zubair Shafiq, Padmini Srinivasan, and Fareed Zaffar. 2021. Through the Looking Glass: Learning to Attribute Synthetic Text Generated by Language Models. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics*.

Priyadarshan Narayanasamy, Swastik Agrawal, Klint Faber, and Fardina Fathmiul Alam. 2026. Beyond Perplexity: Character Distribution Signatures and the MDTA Benchmark for AI Text Detection. *arXiv preprint arXiv:2605.01647*.

Vishakh Padmakumar and He He. 2024. Does Writing with Language Models Reduce Content Diversity? In *Proceedings of the International Conference on Learning Representations*.

Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. 2024. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*.

Krishna Pillutla, Swabha Swayamdipta, Rowan Zellers, John Thickstun, Sean Welleck, Yejin Choi, and Zaid Harchaoui. 2021. MAUVE: Measuring the Gap Between Neural Text and Human Text using Divergence Frontiers. In *Advances in Neural Information Processing Systems*.

Qwen Team. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.

Zixin Rao, Youssef Mohamed, Shang Liu, and Zeyan Liu. 2025. Two Birds with One Stone: Multi-Task Detection and Attribution of LLM-Generated Text. *arXiv preprint arXiv:2508.14190*.

Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. 2025. Can AI-Generated Text be Reliably Detected? Stress Testing AI Text Detectors Under Various Attacks. *Transactions on Machine Learning Research*.

Rylan Schaeffer, Joshua Kazdan, Alvan Caleb Arulandu, and Sanmi Koyejo. 2025. Position: Model Collapse Does Not Mean What You Think. *arXiv preprint arXiv:2503.03150*.

Mohamed El Amine Seddik, Suei-Wen Chen, Soufiane Hayou, Pierre Youssef, and Merouane Debbah. 2024. How Bad is Training on Synthetic Data? A Statistical Analysis of Language Model Collapse. In *Proceedings of the First Conference on Language Modeling (COLM)*.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting Pretraining Data from Large Language Models. In *Proceedings of the International Conference on Learning Representations*.

Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. Deep Neural Networks for Rank-Consistent Ordinal Regression Based On Conditional Probabilities. *Pattern Analysis and Applications*, 26:941–955.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759.

Irene Solaiman, Miles Brundage, Jack Clark, Amanda Askell, Ariel Herbert-Voss, Jeff Wu, Alec Radford, Gretchen Krueger, Jong Wook Kim, Sarah Kreps, Miles McCain, Alex Newhouse, Jason Blazakis, Kris McGuffie, and Jasmine Wang. 2019. Release Strategies and the Social Impacts of Language Models. *arXiv preprint arXiv:1908.09203*.

Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, Valentin Hofmann, Ananya Harsh Jha, Sachin Kumar, Li Lucy, Xinxi Lyu, Nathan Lambert, Ian Magnusson, Jacob Morrison, Niklas Muennighoff, and 17 others. 2024. Dolma: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Efstathios Stamatatos. 2009. A Survey of Modern Authorship Attribution Methods. *Journal of the American Society for Information Science and Technology*, 60(3):538–556.

Carolin Strobl, Anne-Laure Boulesteix, Thomas Kneib, Thomas Augustin, and Achim Zeileis. 2008. Conditional variable importance for random forests. *BMC Bioinformatics*, 9:307.

Eduard Tulchinskii, Kristian Kuznetsov, Laida Kushnareva, Daniil Cherniavskii, Sergey Nikolenko, Evgeny Burnaev, Serguei Barannikov, and Irina Piontkovskaya. 2023. Intrinsic Dimension Estimation for Robust Detection of AI-Generated Texts. In *Advances in Neural Information Processing Systems*.

Adaku Uchendu, Thai Le, Kai Shu, and Dongwon Lee. 2020. Authorship Attribution for Neural Text Generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.

Vivek Verma, Eve Fleisig, Nicholas Tomlin, and Dan Klein. 2024. Ghostbuster: Detecting Text Ghostwritten by Large Language Models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*.

Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Osama Mohammed Afzal, Tarek Mahmoud, Giovanni Puccetti, Thomas Arnold, Chenxi Whitehouse, Alham Fikri Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2024. SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection. In *Proceedings of the International Workshop on Semantic Evaluation*.

Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. 2025. Unveiling Attractor Cycles in Large Language Models: A Dynamical Systems View of

Successive Paraphrasing. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Guillaume Wenzek, Marie-Anne Lachaux, Alexis Conneau, Vishrav Chaudhary, Francisco Guzmán, Armand Joulin, and Edouard Grave. 2020. CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data. In *Proceedings of the Language Resources and Evaluation Conference*.

Alexander Wettig, Aatmik Gupta, Saumya Malik, and Danqi Chen. 2024. QuRating: Selecting High-Quality Data for Training Language Models. In *Proceedings of the International Conference on Machine Learning*.

Junchao Wu, Shu Yang, Runzhe Zhan, Yulin Yuan, Lidia Sam Chao, and Derek Fai Wong. 2025. A Survey on LLM-Generated Text Detection: Necessity, Methods, and Future Directions. *Computational Linguistics*.

Sang Michael Xie, Shibani Santurkar, Tengyu Ma, and Percy Liang. 2023. Data Selection for Language Models via Importance Resampling. In *Advances in Neural Information Processing Systems*.

Xianjun Yang, Wei Cheng, Yue Wu, Linda Petzold, William Yang Wang, and Haifeng Chen. 2024. DNA-GPT: Divergent N-Gram Analysis for Training-Free Detection of GPT-Generated Text. In *Proceedings of the International Conference on Learning Representations*.

Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. Defending Against Neural Fake News. In *Advances in Neural Information Processing Systems*.

Note: Unless otherwise stated, appendix experiments use the original six-generator subset (Pythia-2.8B/1B, OLMo-2-1B, Qwen3-1.7B/0.6B/8B). Main paper results (§4) use the full eight-generator set (adding Llama-3.2-3B and Mistral-7B-v0.3).

a Dataset Statistics

Table 4 reports per-generator sample counts after filtering (removing generation failures, exact duplicates, and passages shorter than 50 whitespace tokens). Under the 3-class evaluation scheme, the dataset contains 7,963 gen-0 samples (21.5%), 7,896 gen-1 (21.3%), and 21,199 gen-2+ (57.2%, combining depths 2–4). The gen-2+ majority reflects the chain structure (three deeper depths per chain vs. one each for gen-0 and gen-1); class-balanced sampling is applied during evaluation.

Generator	Family	Params	Samples
Pythia-1B	Pythia	1.0B	4,555
Pythia-2.8B	Pythia	2.8B	4,609
OLMo-2-1B	OLMo	1.5B	4,713
Qwen3-0.6B	Qwen	0.6B	4,538
Qwen3-1.7B	Qwen	1.7B	4,617
Qwen3-8B	Qwen	8.2B	4,497
Llama-3.2-3B	Llama	3.2B	4,680
Mistral-7B-v0.3	Mistral	7.0B	4,849
Total			**37,058**

Table 4: Per-generator sample counts after filtering (8 generators, 5 families). Each generator starts from $\sim$1,000 shared Wikipedia seeds and produces chains of depth 4 (5 texts per chain). Attrition varies by generator due to differing generation failure rates.[3]

Figure 3 shows the full distribution of each feature across five generation depths (0–4), pooled across all eight generators ($N = 37{,}058$). The violin plots confirm the DDR pattern: drifting features (entropy mean, TTR, hapax ratio) show progressive narrowing and location shift, while shape features (entropy max, tail decay) maintain approximately constant distributions across depths.

b Length Confound Analysis

Surface diversity features (TTR, hapax ratio) mechanically depend on text length: shorter texts yield higher type-token ratios by construction. If generation depth systematically shortens text, observed depth trends could be length artifacts rather than genuine distributional shifts. We investigate with three complementary analyses.

(a) Length–depth correlation. Spearman rank correlation between word count and generation depth is weak across all eight generators ($|\rho| < 0.14$); depth does not systematically shorten or lengthen text in our corpus.

(b) Length–feature correlations. While length–depth correlation is weak, individual features show moderate length dependence: tail_decay ($|\rho| = 0.38$), hapax_ratio ($|\rho| = 0.30$), and TTR ($|\rho| = 0.29$). Reference-model features show weaker length correlations (entropy_mean $|\rho| = 0.09$, entropy_std $|\rho| = 0.18$). This motivates the length-matched evaluation below: if length mediates some depth–feature association, controlling for length will reduce classification performance.

(c) Length-matched classification. We subsample the dataset so that word-count distributions are

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[3] Table 3 evaluates on a subset of 34,131 samples for which all detector scores (FastDetectGPT, Binoculars, LD-Score, Avg-LogProb) are available.

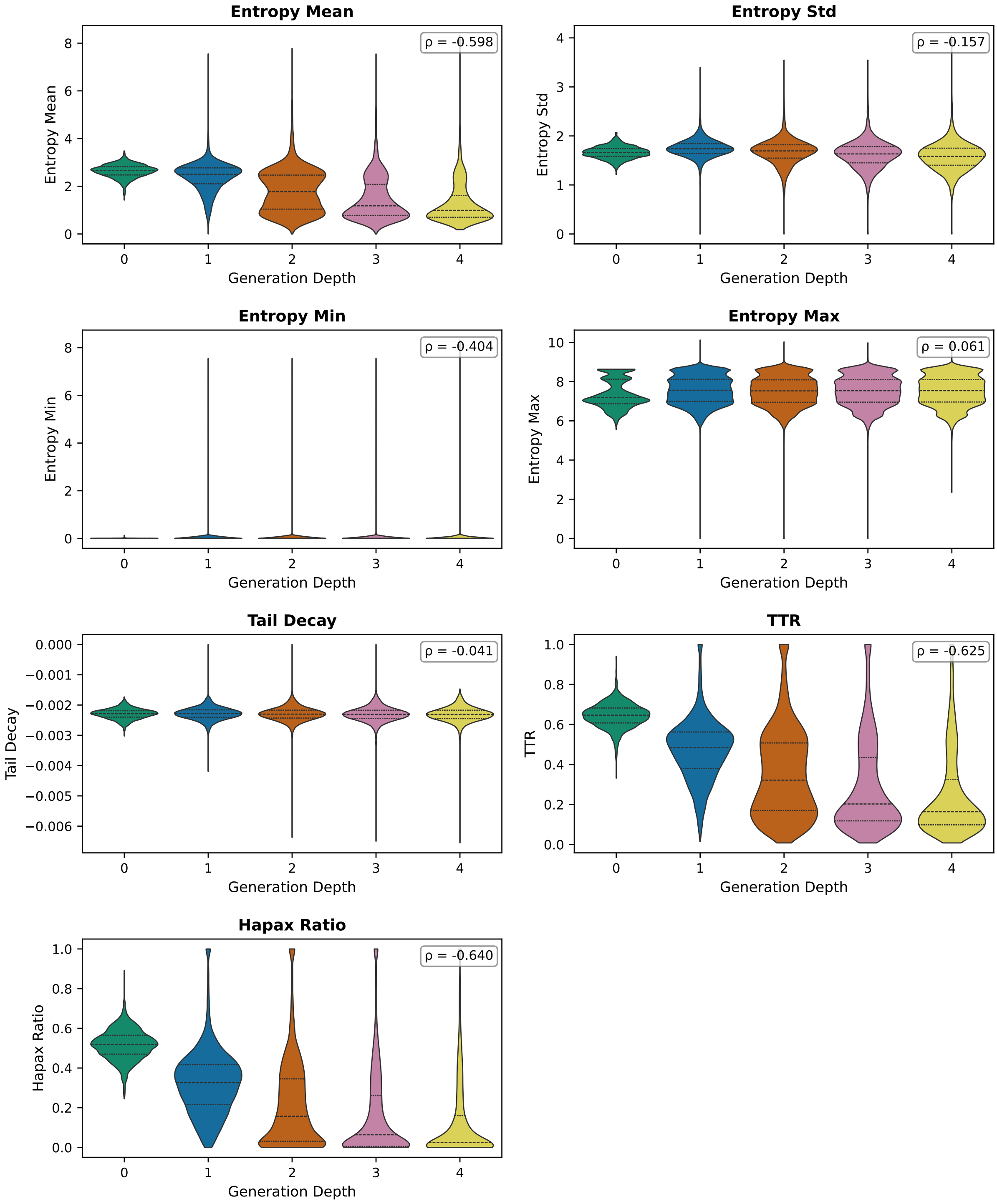

Figure 3: Feature distributions by generation depth (violin plots, $N = 37{,}058$). Drifting features (entropy mean, TTR, hapax ratio: $|\rho| > 0.60$) show clear location shifts and narrowing with increasing depth. Shape features (entropy max, tail decay: $|\rho| < 0.10$) remain approximately stationary. Quartile lines indicate median and interquartile range.

equalized across depth classes (gen-0 / gen-1 / gen-2+), then retrain the binary XGBoost classifier under pooled LOGO (8-fold). Length-matched binary F1 = 0.724, down from 0.770 on the unmatched set ($\Delta = -0.046$). To distinguish this from a sample-size effect, we run a random-subsample control at equal N without length matching: F1 = 0.767, which confirms that the 0.046 drop reflects genuine length deconfounding rather than reduced data.

Conclusion. Length accounts for approximately 6% of discriminative performance ($\Delta = 0.046$ out of 0.770). The majority of the depth signal is length-independent: length-matched F1 (0.724) remains far above chance and significantly above all individual detection baselines (§4.3). Surface features (TTR, hapax_ratio) partially co-vary with length, but their depth-discriminative contribution persists after length control, consistent with generation-induced vocabulary compression operating independently of text truncation.

c Prompt Type Generalization

We evaluate depth profiling under three prompt types using Pythia-2.8B. Each uses 995 seeds × 5 depths. The exact templates are:

- **Continuation**: "`Continue the following text: {text}`"
- **Elaboration**: "`Elaborate on the following text: {text}`"
- **Inspired rewriting**: "`Write a new passage inspired by the following text: {text}`"

Within-prompt-type classification. 5-fold CV macro-F1: continuation 0.78, rewriting 0.74, elaboration 0.73. All three prompt types exhibit the same differential depth response pattern: drifting features (hapax ratio, TTR, entropy mean) show consistent monotonic trends, while near-stationary features (entropy max, tail decay) change by less than 5%.

Cross-prompt-type transfer (LOPTO). Leave-one-prompt-type-out evaluation yields average F1 = 0.753, which confirms that depth signal transfers across prompt types. Feature decay rates differ: continuation shows the strongest drift (16.9% per depth step), followed by rewriting (6.1%) and elaboration (1.4%), consistent with continuation chains applying less semantic constraint on the output.

d Finer-Grained Depth Classification

We evaluate finer-grained 5-class depth classification (gen-0 through gen-4) using the same 7-feature ensemble and LOGO protocol described in §3.

Held-out Generator	Macro F1	Weighted F1	Accuracy
OLMo-2-1B	0.484	0.506	0.524
Pythia-1B	0.543	0.568	0.575
Pythia-2.8B	0.523	0.540	0.560
Qwen3-0.6B	0.513	0.521	0.546
Qwen3-1.7B	0.510	0.536	0.548
Qwen3-8B	0.434	0.455	0.493
Average	**0.501 ± 0.038**	**0.521**	**0.541**

Table 5: 5-class LOGO evaluation. Each row holds out one generator for testing, training on the remaining five.

Metric	Gen-0	Gen-1	Gen-2	Gen-3	Gen-4
Avg F1	0.921	0.634	0.268	0.259	0.423

Table 6: Per-class F1 averaged across all six LOGO folds.

Analysis. Moving from 3-class (gen-0 / gen-1 / gen-2+) to 5-class classification reduces LOGO macro F1 to approximately 0.50 (six-generator subset; the 3-class baseline for this subset is comparable to the 8-generator result of 0.762 in Table 2). The degradation is concentrated in the middle depths: gen-2 and gen-3 achieve only 0.268 and 0.259 F1, respectively. The aggregated confusion matrix reveals that 36.5% of true gen-2 samples are misclassified as gen-1, while 37.8% of true gen-3 samples are predicted as gen-4. Adjacent depths (gen-2↔gen-3, gen-3↔gen-4) account for the majority of errors, consistent with the intuition that deeper recursive generations converge in surface-level statistical signatures. Gen-0 (human text) remains highly distinguishable (F1 = 0.921), and gen-1 retains moderate separability (F1 = 0.634). Shape features that are near-stationary in the 3-class setting become substantially more important under 5-class evaluation: permutation importance of entropy_std increases by 93% and tail_decay by 400% relative to 3-class, yet these gains are insufficient to resolve adjacent depths. These results support the 3-class grouping adopted in the main paper: collapsing gen-2+ preserves the most informative boundary (human vs. first-generation vs. deeper recursion) while avoiding the noise introduced by near-identical adjacent depths.

Classifier comparison. To verify that 5-class degradation reflects feature-level limitations rather than classifier choice, we evaluate four alternatives under the same protocol: HistGradientBoosting (macro F1 = 0.501, 95% bootstrap CI [0.496, 0.506]), random forest (0.476), logistic regression (0.447), and Frank-Hall ordinal reduction (0.451). All classifiers perform comparably to the default XGBoost ensemble (0.501), indicating that classifier capacity is not the bottleneck. The gen-2/gen-3 boundary is the primary failure mode across all methods: these two classes are effectively inseparable with the current feature set. This establishes 3-class granularity as the practical ceiling for surface and entropy features. At the 3-class level, ordinal-specific methods (CORAL ordinal regression, Frank-Hall reduction) similarly fail to improve over multiclass XGBoost (§4), which confirms that the bottleneck is feature separability across all granularities, not the modeling of label ordinality.

e End-to-End Depth Classification (QLoRA)

To assess whether depth signal extends beyond handcrafted features, we fine-tune a pretrained language model end-to-end as a depth classifier using QLoRA (Dettmers et al., 2023).

Setup. We fine-tune Qwen3-0.6B-Base with LoRA rank 16, learning rate 2×10^{-4}, and 3 epochs, framing depth estimation as 5-class sequence classification (gen-0 through gen-4).[4] For 3-class evaluation, we collapse predictions into gen-0 / gen-1 / gen-2+ post hoc. All results use the same LOGO protocol as the main experiments.

Table 7: QLoRA end-to-end depth classification vs. XGBoost on handcrafted features, under LOGO evaluation.

Method	5-class F1	3-class F1	Per-class F1 (5-class)
XGBoost (7 features)	0.501 ± 0.038	0.789	—
QLoRA (Qwen3-0.6B)	0.583 ± 0.010	0.905 ± 0.003	0.999 / 0.796 / 0.413 / 0.259 / 0.447

Results. QLoRA achieves 3-class LOGO F1 = 0.905 (±0.003) under 6-generator LOGO, a +11.6 pp improvement over the same-subset XGBoost baseline (0.789). On 5-class evaluation, QLoRA reaches F1 = 0.583 (±0.010) vs. 0.501 for XGBoost (+8.2 pp), with the largest gains at gen-1 (0.796 vs. 0.634) and gen-2 (0.413 vs. 0.268). The gen-2/gen-3 boundary remains the primary failure mode (gen-3 F1 = 0.259), consistent with the feature-level analysis in §5.

Hyperparameter robustness. A second run with LoRA rank 32 and seed 123 yields 5-class F1 = 0.587 (±0.005) and 3-class F1 = 0.905 (±0.004), which confirms stability across hyperparameter and seed variation ($\Delta < 1$ pp on both metrics).

Implications. The substantial 3-class improvement demonstrates that raw token embeddings contain depth signal beyond what summary statistics capture; the seven handcrafted features are a strong but lossy compression of the full distributional information. End-to-end fine-tuning exploits token-level patterns (e.g., positional regularities, n-gram repetitions) that aggregate features cannot represent. This confirms that the depth signal documented in our main analysis is real and exploitable beyond engineered features, while the handcrafted approach retains practical advantages: interpretability, computational efficiency (no GPU inference), and the ability to reveal the differential depth response pattern that motivates the task formulation.

f Leave-One-Feature-Out Analysis

We complement the permutation importance analysis (§4.2) with a leave-one-feature-out ablation. Each row drops one of the seven features and re-evaluates the full 6-gen LOGO protocol; Δ is the change relative to the full 7-feature ensemble (F1 = 0.789).

Dropped feature	LOGO F1	Δ vs Full
entropy_max	0.774	−1.50 pp
entropy_mean	0.783	−0.63 pp
ttr	0.784	−0.55 pp
hapax_ratio	0.785	−0.42 pp
tail_decay	0.786	−0.33 pp
entropy_std	0.786	−0.29 pp
entropy_min	0.789	+0.01 pp
Full (7 features)	0.789	—

Table 8: Leave-one-feature-out ablation (6-gen LOGO). Each row drops one feature and re-evaluates with the LOGO protocol. Sorted by impact magnitude.

Analysis. Entropy_max causes the largest degradation when removed (−1.5 pp), despite ranking only fourth in permutation importance (split frequency 0.063). This indicates that entropy_max provides unique signal (capturing the most surprising token under the reference model) that is not redundant with other features. Conversely, entropy_min is fully redundant ($\Delta = +0.01$ pp), consistent with its near-zero permutation importance (0.015). Six of seven features contribute positively to the ensemble, which confirms that the 7-feature design is not over-specified.

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[4] Evaluated on the original 6-generator subset; XGBoost 3-class F1 = 0.789. We expect 8-generator training to further improve given additional diversity; the comparison direction is preserved.

f.1 Candidate Feature Screening

To verify that the seven retained features are not an arbitrary subset, we screened six additional NLL-distributional statistics as candidate features: median, 90th percentile (p90), interquartile range (IQR), coefficient of variation (CV), skewness, and kurtosis. Table 9 reports their DDR coordinates, classification impact, and correlation with retained features.

Table 9: Candidate feature screening. All six candidates fall in the drifting DDR zone and correlate strongly ($|r| > 0.65$) with existing features. Adding any combination to the baseline degrades or leaves unchanged LOGO 3-class F1; LOFO $\Delta > 0$ indicates that removing the candidate *improves* classification.

Candidate	$\lvert\rho\rvert$	d	**LOFO ΔF1**	$\lvert r\rvert_{\max}$	**Most correlated**
median	0.685	0.794	+0.008	0.960	entropy_mean
p90	0.693	0.960	+0.000	0.937	entropy_mean
IQR	0.708	1.079	+0.007	0.908	entropy_std
CV	0.633	0.980	−0.000	0.761	entropy_mean
skewness	0.651	0.940	+0.010	0.780	entropy_mean
kurtosis	0.635	0.769	−0.003	0.650	entropy_mean

All six candidates occupy the drifting zone ($|\rho| > 0.63$, $d > 0.77$), overlapping with entropy mean, TTR, and hapax ratio; none introduces a new axis of depth-dependent variation. Three candidates—median (r=0.96), p90 (r=0.94), and IQR (r=0.91)—are near-perfect linear proxies of existing features. Adding all six to the 6-feature baseline[5] reduces 3-class LOGO F1 from 0.733 to 0.722, consistent with overfitting from redundant inputs. LOFO analysis on the extended 12-feature set confirms that removing skewness (+0.010), median (+0.008), or IQR (+0.007) *improves* classification, indicating these features inject noise rather than signal. The retained seven features span three complementary DDR zones (drifting, intermediate, near-stationary) and two modalities (reference-model NLL, surface lexical diversity); additional distributional statistics provide no incremental information.

g Confusion Matrix

Table 10 reports the aggregated confusion matrix from the original six-generator subset (Pythia-2.8B, Pythia-1B, OLMo-2-1B, Qwen3-1.7B, Qwen3-0.6B, Qwen3-8B) under leave-one-generator-out (LOGO) evaluation, collapsed to three ordinal classes: gen-0 (human), gen-1 (first-generation), and gen-2+ (depth ≥ 2). The later-added Llama-3.2-3B and Mistral-7B-v0.3 are evaluated in the full 8-generator LOGO results (Table 2).

	Pred gen-0	**Pred gen-1**	**Pred gen-2+**
True gen-0	5,937 (99.4%)	35 (0.6%)	0 (0.0%)
True gen-1	368 (6.2%)	4,437 (75.0%)	1,113 (18.8%)
True gen-2+	464 (3.0%)	3,646 (23.3%)	11,529 (73.7%)

Table 10: Aggregated 3-class confusion matrix from 6-generator LOGO evaluation ($N = 27{,}529$). Each cell shows count and row-normalized percentage. Gen-0 is near-perfectly identified. Gen-1 errors are asymmetric: misclassification as gen-2+ (18.8%) is 3× more frequent than as gen-0 (6.2%). Gen-1 text already carries detectable recursive distortion.

Figure 4 provides a heatmap visualization of the confusion matrix. The row-normalized view highlights near-perfect gen-0 classification (99.4% recall), gen-1 confusion primarily with gen-2+ (18.8%) rather than gen-0 (6.2%), and gen-2+ misclassification predominantly into gen-1 (23.3%), consistent with DDR predictions.

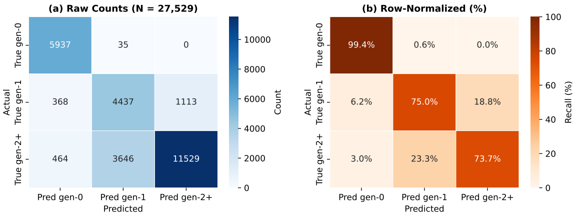

Figure 4: 3-class LOGO confusion matrix ($N = 27{,}529$). (a) Raw counts. (b) Row-normalized recall percentages. Gen-0 achieves near-perfect classification; the primary confusion axis is gen-1 ↔ gen-2+.

h Classifier Sensitivity

To verify that the depth signal resides in the features rather than the classifier, we repeat the 6-generator LOGO evaluation using Logistic Regression and Random Forest in place of XGBoost (Table 11). All other settings (the same 7 features, 3-class labels, and class-balanced downsampling) are kept identical. For Logistic Regression we apply standard scaling to the features.

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[5] Tail decay requires full-vocabulary logits unavailable in the token-NLL subset used for this analysis; we report 6-feature baselines for fair comparison.

Table 11: LOGO macro-F1 across three classifiers. Each column shows the held-out generator F1. (6-generator LOGO; 3-class.)

Classifier	Avg F1	OLMo-2	Py-1B	Py-2.8B	Qw-0.6B	Qw-1.7B	Qw-8B
XGBoost	0.789	0.764	0.856	0.804	0.824	0.806	0.683
Random Forest	0.790	0.764	0.856	0.804	0.825	0.807	0.684
Logistic Regr.	0.725	0.719	0.781	0.740	0.745	0.743	0.624

Both tree-based classifiers achieve LOGO F1 $\approx$ 0.79, and even a linear model reaches 0.725, well above the 0.33 chance baseline for three classes. The per-generator ranking is also consistent across classifiers (Pythia-1B easiest, Qwen3-8B hardest), confirming that discriminative power originates in the entropy and lexical features, not in classifier capacity. Ordinal-specific methods (CORAL (Cao et al., 2020), Frank-Hall (Frank and Hall, 2001)) achieve comparable F1 with overlapping CIs, which confirms that 3-class granularity provides insufficient resolution for ranking-aware losses to improve over standard multiclass classification.

Sampling Sensitivity Analysis

To assess robustness to decoding temperature, we regenerate continuation chains at $t \in \{0.5, 0.9\}$ with fixed nucleus p=0.9 (matching the default) for four generators spanning three model families: Pythia-2.8B, OLMo-2-1B, Qwen3-1.7B, and Qwen3-0.6B (Table 12).

Table 12: Depth classification under varying temperature (fixed top-p = 0.9). Binary: gen-1 vs. gen-2+. LOGO evaluation; 95% bootstrap CIs.*†

Model	Temp	LOGO Binary F1	95% CI	ρ(ent. mean)
Pythia-2.8B†	0.5	**0.814**	[.798, .828]	−0.58
Pythia-2.8B†	0.7	0.740	[.725, .755]	−0.60
Pythia-2.8B†	0.9	0.480	[.465, .498]	−0.30
OLMo-2-1B	0.5	**0.783**	[.768, .798]	−0.72
OLMo-2-1B	0.7	0.695	[.679, .711]	−0.60
OLMo-2-1B	0.9	0.453	[.437, .469]	−0.26
Qwen3-1.7B†	0.5	0.751	—	−0.60
Qwen3-1.7B†	0.7	0.751	—	−0.60
Qwen3-1.7B†	0.9	0.566	—	−0.37
Qwen3-0.6B	0.5	0.711	[.694, .729]	−0.67
Qwen3-0.6B	0.7	0.691	[.673, .709]	−0.60
Qwen3-0.6B	0.9	0.598	[.582, .615]	−0.29

[*] OLMo temperature experiments use OLMo-2-0425-1B (April 2025 release); the original OLMo-2-1B-hf weights are no longer publicly available.

[†] Pythia-2.8B and Qwen3-1.7B use 6-generator LOGO (pre-expansion); OLMo-2-1B and Qwen3-0.6B use 8-generator LOGO. Within-model comparisons are unaffected.

Temperature sensitivity is asymmetric and model-dependent. High-temperature degradation (t=0.9) is universal: all four generators lose 9.3–26.0 pp in binary F1, with entropy mean's $|\rho|$ weakening to 0.26–0.37 (vs. 0.58–0.72 at t=0.5). Low-temperature improvement (t=0.5) is model-dependent: Pythia-2.8B and OLMo-2-1B gain +7.4 and +8.8 pp respectively, while both Qwen models show negligible change (0–2.0 pp). This split tracks model family rather than scale: Qwen3-0.6B (smallest) and Qwen3-1.7B both plateau, while OLMo-2-1B (similar scale to Qwen3-1.7B) benefits substantially. One possible explanation is that Qwen models produce more peaked output distributions at t=0.7, leaving less room for low-temperature compression to sharpen the depth signal.

The method operates reliably within $t \in [0.5, 0.7]$ across all tested models. Degradation onset at t=0.9 defines the practical upper bound, consistent with the hypothesis that elevated temperature injects sufficient randomness to flatten the distributional drift that DDR exploits (§5).

Per-Generator Performance

Figure 5 visualizes per-generator LOGO performance. Pythia models (1B, 2.8B) achieve highest F1, while larger models (Mistral-7B, Qwen3-8B) show degradation. The scale effect (larger models → lower F1) is consistent with higher-capacity models generating text statistically closer to human writing.

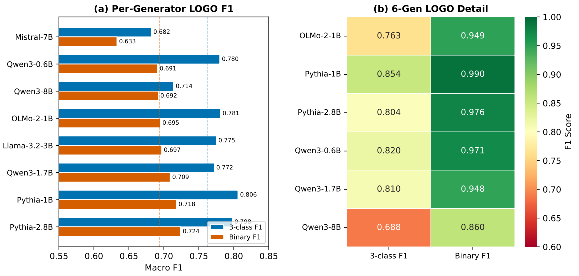

Figure 5: Per-generator LOGO performance. (a) 8-generator binary and 3-class F1, sorted by binary F1. Dashed lines: pool average. (b) 6-generator LOGO detail heatmap showing the performance spread across model families.

i Robustness to Text Perturbation

We assess robustness to post-generation text modification by randomly replacing words at varying rates and re-evaluating the 8-generator LOGO pipeline (Table 13). Random word replacement represents a worst-case perturbation: real human editing preserves semantic coherence, so practical degradation is expected to be substantially lower than these estimates.

Table 13: Robustness to random word replacement under 8-gen LOGO evaluation (binary: gen-1 vs. gen-2+). Bootstrap 95% CIs on binary F1 ($n = 1000$).

Perturbation rate	Binary F1	95% CI	3-class F1
0% (baseline)	0.695	[0.680, 0.710]	0.737
5%	0.650	[0.635, 0.666]	0.641
10%	0.614	[0.599, 0.629]	0.492
20%	0.591	[0.576, 0.605]	0.367

Binary depth discrimination (gen-1 vs. gen-2+) degrades sublinearly: the first 5% replacement causes the largest drop (−4.6 pp), with diminishing marginal impact thereafter (−3.6 pp for 5%→10%, −2.3 pp for 10%→20%). At 10% replacement, binary F1 remains 0.614, well above the 0.50 random baseline, which indicates that depth signal survives moderate perturbation.

Three-class classification is more sensitive (0.737→0.367 at 20%), because distinguishing gen-0 from gen-1 relies heavily on lexical diversity features (TTR, hapax ratio) that are directly disrupted by word replacement. The binary metric, which excludes gen-0, is inherently more robust since it depends primarily on reference-model entropy features distributed across many tokens.

Feature subset degradation. Table 14 compares perturbation robustness of the full 7-feature ensemble against an entropy-only subset (5 reference-model features). The higher unperturbed binary F1 of the entropy-only subset (0.833 vs. 0.695) reflects the strong discriminability of reference-model features for the coarse binary task (gen-1 vs. gen-2+); however, lexical features are essential for the finer-grained ordinal task (3-class F1: 0.737 vs. 0.688) and provide greater robustness under perturbation. At 20% replacement, the 7-feature ensemble surpasses entropy-only on binary F1 (0.591 vs. 0.569), despite starting from a lower baseline. Random word replacement drastically alters per-token context and makes entropy features (computed over the full token sequence) extremely sensitive to noise. Lexical features (TTR, hapax ratio), as document-level summary statistics, are more stable under per-word perturbation and act as stabilizers in the full ensemble. The crossover confirms that degradation in the full feature set reflects genuine signal disruption rather than a lexical-feature artifact.

Baseline comparison. Table 15 compares robustness across detection methods under identical perturbation conditions (8-gen LOGO, binary: gen-1 vs. gen-2+).

Table 14: Feature subset degradation under random word replacement (8-gen LOGO). Entropy-only: 5 reference-model features; 7-feature: full ensemble including lexical features.

Perturb.	Ent-only Bin	7-feat Bin	Ent-only 3-cl	7-feat 3-cl
0%	0.833	0.695	0.688	0.737
5%	0.738	0.650	0.539	0.641
10%	0.667	0.614	0.427	0.492
20%	0.569	0.591	0.314	0.367

Table 15: Baseline robustness under random word replacement (8-gen LOGO, binary F1). Bold indicates highest F1 at each perturbation rate.

Method	0%	5%	10%	20%
7-feature ensemble	**0.695**	0.650	0.614	0.591
LD-Score	0.677	**0.673**	**0.677**	**0.671**
Avg-LogProb	0.682	0.419	0.335	0.375
FastDetectGPT	0.571	0.437	0.468	0.514
Binoculars[†]	0.500	0.461	0.446	0.425

[†] Binoculars operates at chance level on this binary task; its degradation figures are not meaningful.

The 7-feature ensemble achieves the highest unperturbed binary F1 (0.695), but LD-Score surpasses it under perturbation due to near-zero degradation (−0.6 pp across the full 0%→20% range). FastDetectGPT and Avg-LogProb exhibit non-monotonic degradation at higher perturbation rates, likely because severe perturbation introduces distributional artifacts that these token-level methods detect as anomalous. Avg-LogProb is the most fragile overall (−34.7 pp at 10%), while FastDetectGPT drops sharply at 5% (−13.4 pp) before partially recovering. LD-Score is nearly invariant to perturbation, consistent with its reliance on corpus-level distributional divergence rather than per-token likelihoods. The 7-feature ensemble offers the highest starting performance among single-reference methods, with sublinear degradation (−10.4 pp at 20%).

LLM-simulated sentence regeneration. To test robustness under semantically coherent editing, we use Qwen3-0.6B-Base to randomly regenerate 1–3 sentences per passage (avg 1.69 sentences, ∼50% edit distance) and re-evaluate the 8-gen LOGO pipeline. Binary F1 drops from 0.692 to 0.647 (−4.5 pp, 6.5% relative); 3-class F1 drops from 0.739 to 0.667 (−7.1 pp, 9.7% relative). The binary degradation (−4.5 pp) is nearly identical to that caused by 5% random word replacement (−4.6 pp), despite the LLM rephrase altering ∼50% of characters. This asymmetry arises because LLM-generated replacements preserve the underlying statistical signature of the generation process (entropy dynamics, depth-accumulated distributional drift), whereas random word substitution directly destroys local statistical signals. Depth profiling thus captures deep generative structure that survives semantic-preserving edits.

Seed complexity. Performance is stable across seed text complexity: stratifying seeds by vocabulary richness (TTR terciles) yields maximum F1 variation of 1.1 pp across groups.

j Cross-Domain Generalization

We evaluate whether depth signal generalizes beyond Wikipedia by constructing generation chains from three additional domains: News articles, Academic abstracts, and Code documentation. For each domain, we sample 200 seed texts and generate chains of depth 3 using three generators (Pythia-1B, Qwen3-1.7B, Mistral-7B). The same 7-feature ensemble and LOGO protocol are applied.

Important caveat: Cross-domain generation used t=1.0, top-p=1.0, which falls outside the method's reliable operating range ($t \leq 0.7$; Appendix h). Domain and temperature effects are therefore confounded in the results below; the cross-domain signal may partly reflect temperature-induced distributional divergence rather than domain generalization alone.

Table 16: Cross-domain depth classification under LOGO evaluation (binary: gen-1 vs. gen-2+). Three generators × 200 seeds per domain. Bootstrap 95% CIs ($n = 1000$).

Domain	Binary F1	95% CI	3-class F1
Code	0.754	[0.737, 0.770]	0.634
News	0.746	[0.726, 0.764]	0.652
Academic	0.739	[0.720, 0.757]	0.622

All three non-Wikipedia domains achieve binary F1 > 0.73 under leave-one-generator-out evaluation, which demonstrates that depth signal is not an artifact of Wikipedia text properties.

Domain × temperature interaction. Cross-domain generation used t=1.0, top-p=1.0 (vs. the main experiment's t=0.7, top-p=0.9). Supplementary evaluation at t=0.7 reveals a domain × temperature interaction: the main Wikipedia experiment favors lower temperature (depth signal degrades monotonically with rising temperature; Appendix h), while cross-domain evaluation favors higher temperature (t=1.0 yields +3 to +11 pp over t=0.7; News 0.746 vs. 0.712, Academic 0.739 vs. 0.628, Code 0.754 vs. 0.687). This asymmetry likely reflects reference-model calibration: Pythia-410M, trained predominantly on web text, detects subtle per-step drift in Wikipedia but requires larger distributional divergence (produced by higher temperature) to detect depth signal in out-of-distribution domains. This finding suggests that reference model selection is a key factor for cross-domain deployment of depth profiling.

3-class inversion as DDR evidence. In the cross-domain setting, 3-class F1 (0.62–0.65) falls below binary F1 (0.74–0.75), reversing the pattern observed in the main experiment (3-class 0.762 > binary 0.694). This inversion reflects degraded gen-0 vs. gen-1 separation: the Pythia-410M reference model is calibrated to Wikipedia text, so its baseline entropy estimates are less discriminative for out-of-domain human text and compresses the feature distance between gen-0 and gen-1. The binary task (gen-1 vs. gen-2+) is unaffected because both classes are generated under identical domain and decoding conditions; recursive drift is a relative change that does not depend on absolute calibration. This pattern provides additional evidence that DDR captures base-model generation dynamics rather than domain-specific text properties: depth signal remains robust even when the human-vs-machine boundary becomes less distinct.

k Instruction-Tuned Model Analysis

Instruction-Tuned Model Results

Table 17 summarizes depth classification under instruction tuning for three model families. All results use within-model 5-fold CV on single-generator chains; LOGO evaluation is available for two models but uses earlier generator subsets (5–6 generators), so cross-model numbers are not directly comparable across rows.

Table 17: Depth classification: base vs. instruction-tuned models (within-model 5-fold CV). ρ: Spearman correlation of entropy mean with depth. Rev.: number of features (out of 7) whose ρ reverses sign under instruction tuning.

	3-class F1		**Binary F1**		**ρ(ent. mean)**		
Model	Base	Instruct	Base	Instruct	Base	Instruct	**Rev.**
OLMo-2-1B	0.754	0.575	0.845	0.813	−0.64	+0.48	6/7
Qwen3-1.7B	0.769	0.732	0.861	0.875	−0.60	+0.48	6/7
Llama-3.2-3B	0.742	0.544	0.849	0.838	−0.63	+0.23	3/7

Instruction tuning degrades 3-class F1 by 3.7–

19.9 pp across all three families, while binary F1 (gen-1 vs. gen-2+) changes remain small ($|\Delta| \leq 3.2$ pp). The asymmetry indicates that instruction tuning primarily disrupts the gen-0/gen-1 boundary (human vs. first-generation), consistent with RLHF shifting the output distribution away from human-like text. Entropy-mean ρ reverses sign for all three models (-0.60–-0.64 base $\rightarrow$ $+0.23$–$+0.48$ instruct), and 6 of 7 features reverse for OLMo and Qwen vs. 3 of 7 for Llama. This pattern is consistent with instruction tuning shifting the generation attractor rather than eliminating iterative dynamics (§5).

Retraining on instruction-tuned data. To test whether the depth signal persists under instruction tuning with reversed polarity, we retrain the XGBoost classifier on instruction-tuned feature data (Table 18).

Table 18: Classification recovery under instruction-tuned retraining (binary-only: gen-1 vs. gen-2+, gen-0 excluded from training and evaluation). Base→Inst: base-trained classifier tested on instruct data. Note: Table 17 reports higher instruct binary F1 (e.g., OLMo 0.813) because it uses 3-class training with binary evaluation, where gen-0 in training helps the classifier learn the feature space.

Model	Base→Inst	Retrained	Δ
OLMo-2-1B-Inst	0.321	0.590	+0.269
Qwen3-1.7B-Inst	0.500	0.660	+0.160
Qwen3-0.6B-Inst	0.532	0.600	+0.068

Retraining recovers substantial within-model performance (+6.8–26.9 pp), but the retrained F1 (0.590–0.660) remains below base-model levels (0.69–0.72). Three-generator instruct LOGO yields binary F1 = 0.487, indicating weak cross-generator transfer under instruction tuning.

DDR structure under instruction tuning. $k{=}2$ clustering in $(|\rho|, d)$ space on pooled instruct features yields silhouette = 0.372 (vs. 0.660 for base models), indicating that the clean drifting/non-drifting partition is substantially compressed. The compression is model-dependent: OLMo-Instruct shows near-zero $|\rho|$ for TTR and hapax (0.006, 0.025), while Qwen instruct models retain moderate correlations ($|\rho|$=0.48–0.53). Cross-condition transfer is near chance in both directions (base→instruct binary F1 = 0.490; instruct→base F1 = 0.454), with the asymmetry (-0.036) suggesting that instruct-trained features generalize even less to base distributions than the reverse. These results indicate that instruction tuning compresses the feature-space structure underlying DDR rather than simply reversing feature directions (Figure 6).

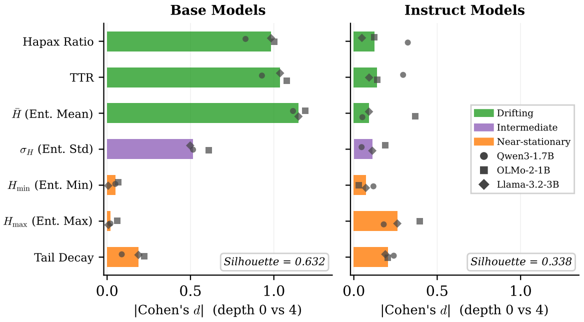

Figure 6: DDR partition under base (left) vs. instruction-tuned (right) models. Bars show pooled $|d|$; markers show individual generators. Base models exhibit a clean drifting/non-drifting separation (silhouette = 0.660); instruction tuning compresses all features toward zero, collapsing the partition (silhouette = 0.372). Entropy max shifts from near-stationary to intermediate under instruct, while formerly drifting features (hapax, TTR) lose most of their effect size.

Reference model sensitivity. One hypothesis for the instruct degradation is reference-model mismatch: the base Pythia-410M reference may produce systematically biased NLL estimates on instruction-tuned outputs, masking depth-dependent variation. To test this, we replace the reference with Qwen3-0.6B-Instruct and recompute all features on the two instruct generators. The hypothesis is partially rejected: Qwen3-1.7B-Instruct shows mixed changes (entropy max d: 0.51 $\rightarrow$ 0.08, becoming near-stationary; entropy std d: 1.07 $\rightarrow$ 1.51, increasing), while OLMo-Instruct loses nearly all depth signal (entropy mean d: 1.33 $\rightarrow$ 0.02; tail decay d: 0.60 $\rightarrow$ 0.06). The model-dependent heterogeneity rules out reference mismatch as a universal explanation for instruct DDR degradation, supporting the interpretation that alignment fundamentally reshapes the generation dynamics rather than introducing a correctable measurement artifact.

End-to-end instruct classification. To test whether depth signal persists at the token level despite summary-statistic degradation, we apply the QLoRA end-to-end classifier (Appendix e) to instruction-tuned data using within-model 5-fold stratified CV (Table 19).

The XGBoost-to-QLoRA gap on the same instruct data is substantial (+17.5 to +31.3 pp), indi-

Table 19: End-to-end QLoRA vs. XGBoost on instruction-tuned data (within-model 5-fold CV, Qwen3-0.6B backbone). Base QLoRA rows shown for reference only; instruct sets ($N\approx4.4$–5.0K) are approximately twice the base sets ($N\approx2.3$K), so cross-condition comparison is confounded by training set size.

	3-class F1			
Model	XGB	QLoRA	**Δ**	N
OLMo-2-1B-Inst	0.575	0.888 ± 0.007	+31.3 pp	4,429
Qwen3-1.7B-Inst	0.732	0.907 ± 0.008	+17.5 pp	4,963
OLMo-2-1B (base)		0.859 ± 0.004		2,379
Qwen3-1.7B (base)		0.904 ± 0.009		2,291

cating that depth signal persists at the token level under instruction tuning but is inaccessible to the seven summary statistics; the degradation reported in Table 17 reflects a feature-level limitation rather than signal absence. Note that instruct samples are approximately twice the base count used here (Table 19), so the instruct-vs-base QLoRA comparison is confounded by training set size; the meaningful comparison is QLoRA-vs-XGBoost on the same instruct data. All results use within-model 5-fold CV, not cross-generator LOGO, limiting generalization claims.

l Error Analysis of Binary Classification

We analyze the distribution and characteristics of binary classification errors (gen-1 vs. gen-2+) across the 8-generator LOGO evaluation ($N = 29{,}095$ samples, overall error rate 28.08%).

Error asymmetry. Errors are heavily asymmetric: 86.2% are false negatives (gen-2+ predicted as gen-1) and only 13.8% are false positives (gen-1 predicted as gen-2+). The classifier systematically under-predicts depth, consistent with the DDR pattern: shallow gen-2+ samples that have not yet accumulated sufficient distributional drift are indistinguishable from gen-1.

Per-depth error rates. Error rates decrease monotonically with depth: depth-2 samples are the primary failure point (48.42% error rate), followed by depth-3 (29.17%) and depth-4 (20.32%). Depth-1 false positive rate is 7.11%. The monotonic decrease confirms that deeper generations accumulate progressively stronger depth signatures; the gen-1→gen-2 transition is therefore the hardest to detect.

Per-generator variation. Table 20 shows error rates vary substantially across generators, correlating with model scale: larger models (Mistral-7B, Qwen3-8B) produce text closer to gen-1 characteristics, which makes depth discrimination harder.

Table 20: Binary classification error rates by generator (8-gen LOGO). Sorted by error rate descending.

Generator	**Family**	**Samples**	**Errors**	**Error rate**
Mistral-7B-v0.3	Mistral	3,853	1,734	45.00%
Qwen3-8B	Qwen	3,502	1,197	34.18%
Llama-3.2-3B	Llama	3,685	1,112	30.18%
OLMo-2-1B	OLMo	3,718	1,054	28.35%
Pythia-2.8B	Pythia	3,614	866	23.96%
Qwen3-1.7B	Qwen	3,621	814	22.48%
Qwen3-0.6B	Qwen	3,542	745	21.03%
Pythia-1B	Pythia	3,560	648	18.20%

Feature-level explanation. Mann-Whitney U tests comparing correctly vs. incorrectly classified samples reveal that drifting features show significant distributional differences between correct and incorrect predictions: entropy_mean ($p < 10^{-300}$), TTR ($p < 10^{-300}$), and hapax_ratio ($p < 10^{-300}$). In contrast, shape features show no discriminating power: entropy_max ($p = 0.584$) and tail_decay ($p = 0.653$). Misclassified gen-2+ samples have systematically higher TTR (0.464 vs. 0.302 for correct) and entropy_mean (2.26 vs. 1.57); these samples retain gen-1-like lexical diversity and have not undergone the distributional convergence characteristic of deeper generations. This is consistent with the DDR pattern (§4.2): errors concentrate where depth-accumulated drift is insufficient, and shape features (which are near-stationary by design) provide no additional discriminative signal for error correction.

m Baseline Comparison Visualization

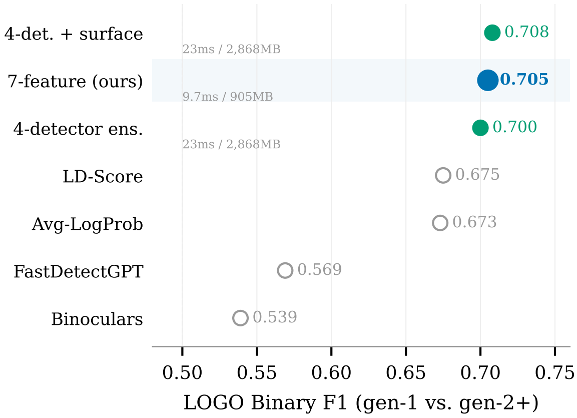

Figure 7: LOGO binary F1 vs. inference cost in ms/text (8-gen pooled). The 7-feature ensemble (one 410M reference model) matches multi-detector ensembles (two models, 1.4B total) at 2.4× lower cost; individual detectors vary widely when repurposed for depth. Inference costs measured on a single NVIDIA RTX 5090.

n Downstream Utility

Can depth-aware data curation improve downstream language model quality? We fine-tune OLMo-2-0425-1B on five data variants (3 seeds each) and evaluate perplexity on a held-out test set (Table 21).

Variant	n	**Test PPL**
All depths	12,299	2.335 ± 0.000
Size-matched (random)	3,951	2.367 ± 0.000
Depth 0–1 only	3,951	2.387 ± 0.001
Binary-filtered	3,951	2.389 ± 0.000
Depth 0 only	797	2.681 ± 0.001

Table 21: Downstream fine-tuning perplexity (OLMo-2-0425-1B, mean ± std across 3 seeds). At equal sample count, random subsampling (2.367) outperforms depth-aware curation (depth 0–1: 2.387, binary-filtered: 2.389).

The results are unambiguous: data quantity dominates depth composition. Training on all 12,299 samples yields the lowest perplexity (2.335). At equal sample count ($n = 3{,}951$), random subsampling (2.367) *outperforms* both depth-stratified selection (depth 0–1: 2.387) and binary detection filtering (2.389). Cross-seed variance is below 0.002 across all variants, which confirms the robustness of this null result.

Restricting training to human-only text (depth 0, $n = 797$) produces a sharp perplexity increase to 2.681, driven primarily by the 5× reduction in data volume rather than depth composition per se.

Equivalence testing. We apply a two one-sided t-test (TOST) with a $\pm 2\%$ relative margin (equivalence bounds ± 0.047 on the ppl scale) to formally assess whether the depth-aware and random subsampling conditions produce practically indistinguishable perplexity. At equal sample count ($n = 3{,}951$): size-matched (random) ppl = 2.367 vs. depth 0–1 ppl = 2.387, $\Delta = 0.020$. TOST confirms equivalence ($p = 1.1 \times 10^{-4}$); the two curation strategies yield practically indistinguishable downstream quality within the $\pm 2\%$ margin. Cross-seed standard deviation is below 0.001 for both variants.

Comparing the full dataset ($n = 12{,}299$, ppl = 2.335) against the depth-filtered subset ($n = 3{,}951$, ppl = 2.387): TOST *rejects* equivalence (minimum equivalence margin = 0.053, exceeding the $\pm 2\%$ bound), which confirms that data quantity drives perplexity reduction beyond what depth-aware selection can recover.

Knowing the generation depth of a document does not, by itself, make it more or less useful for training; at fixed sample count, depth-aware curation is statistically equivalent to uninformed random subsampling.

Margin Sensitivity Analysis

To assess the robustness of the equivalence conclusion to margin choice, we repeat the TOST procedure across multiple equivalence margins (Lakens, 2017), following the recommendation to report sensitivity across plausible bounds rather than relying on a single threshold (Table 22).

Comparison	**Margin**	**Equiv?**	**TOST p**
Full (12,299) vs. Random (3,951)	$\pm 1\%$	No	1.0
	$\pm 2\%$	Yes	1×10^{-6}
	$\pm 3\%$	Yes	$< 10^{-10}$
	$\pm 5\%$	Yes	$< 10^{-10}$
Full (12,299) vs. Depth 0–1 (3,951)	$\pm 1\%$	No	1.0
	$\pm 2\%$	No	1.0
	$\pm 3\%$	Yes	$< 10^{-10}$
	$\pm 5\%$	Yes	$< 10^{-10}$

Table 22: TOST equivalence testing across multiple margins. “Equiv?” indicates whether equivalence is established at $\alpha = 0.05$.

The full dataset vs. random subsample comparison passes equivalence at the $\pm 2\%$ margin ($p = 10^{-6}$), while the full vs. depth 0–1 comparison requires a $\pm 3\%$ margin. In absolute terms, perplexity differences are small: $\Delta = 0.032$ (full vs. random, 1.4% of baseline) and $\Delta = 0.052$ (full vs. depth 0–1, 2.2% of baseline).

Cohen’s d values are 86.0 (full vs. random) and

119.7 (full vs. depth 0–1). These appear large but are unreliable: with only $n = 3$ seeds per condition, the pooled standard deviation is extremely small (<0.001), artificially inflating d. The inflated effect sizes reflect low seed-level variance (confirming reproducibility) rather than practically meaningful perplexity differences. The TOST framework with pre-specified relative margins provides a more appropriate assessment than Cohen's d in this low-seed regime. However, with only n=3 seeds per condition, the TOST results have limited generalizability: the test is valid given observed variance but the conclusion generalizes over only three random seeds. We characterize this as a preliminary observation rather than a definitive null result.

o Cross-Model Chain Per-Class Detail

This appendix reports per-generator breakdowns for the cross-model chain evaluation described in §4.1.

Class	Prec.	Rec.	F1	95% CI
gen-0	0.921	0.948	0.935	[.923, .946]
gen-1	0.737	0.803	0.768	[.748, .788]
gen-2+	0.842	0.742	0.789	[.768, .809]
Macro	—	—	**0.831**	[.817, .844]

Table 23: Cross-model chain per-class performance (generator-level leave-one-out, $N = 2{,}964$, $n_{\text{generators}} = 5$: Pythia-2.8B/1B, OLMo-2-1B, Qwen3-0.6B/1.7B). Macro F1 = 0.831 under class balancing; under natural class distribution, macro F1 = 0.789. Binary F1 (gen-1 vs. gen-2+) = 0.830 (95% CI [.818, .841], case-level bootstrap, $n = 2{,}000$). Cross-model chains avoid the gen-0 seed sharing issue present in single-generator LOGO.

p Feature Distribution Analysis

DDR Trend Lines

Figure 8 shows the per-feature depth trend (mean across all samples at each depth) with individual generator traces overlaid. The consistent downward trajectories for drifting features across all generators confirm that DDR is a universal property of recursive generation. Shape features exhibit flat trajectories regardless of generator, with near-zero R^2 for the linear fit. R^2 values are computed from ordinary least-squares regression on depth-level means (five points per generator, one per depth 0–4), not per-sample regression; the annotated Spearman ρ provides the complementary per-sample monotonicity measure.

UMAP Embedding

Figure 9 shows a 2D UMAP projection of the 7-feature space (7,500 stratified samples, StandardScaler normalization, $k = 30$ neighbors, min_dist = 0.3). Panel (a) reveals a depth gradient from gen-0 through gen-4, with progressive overlap at deeper generations consistent with the convergence that motivates the gen-2+ class collapse. Panel (b) shows the 3-class view: gen-0 forms a separable cluster, while gen-1 and gen-2+ overlap substantially, explaining the classification difficulty at this boundary.

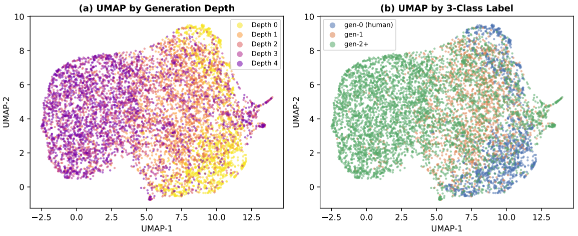

Figure 9: UMAP 2D projection of the 7-feature space ($N = 7{,}500$ stratified sample). (a) Colored by raw generation depth (0–4): a gradient structure emerges, with deeper depths converging. (b) Colored by 3-class label: gen-0 separates clearly; gen-1 and gen-2+ overlap, consistent with the high false-negative rate at the gen-1/gen-2+ boundary.

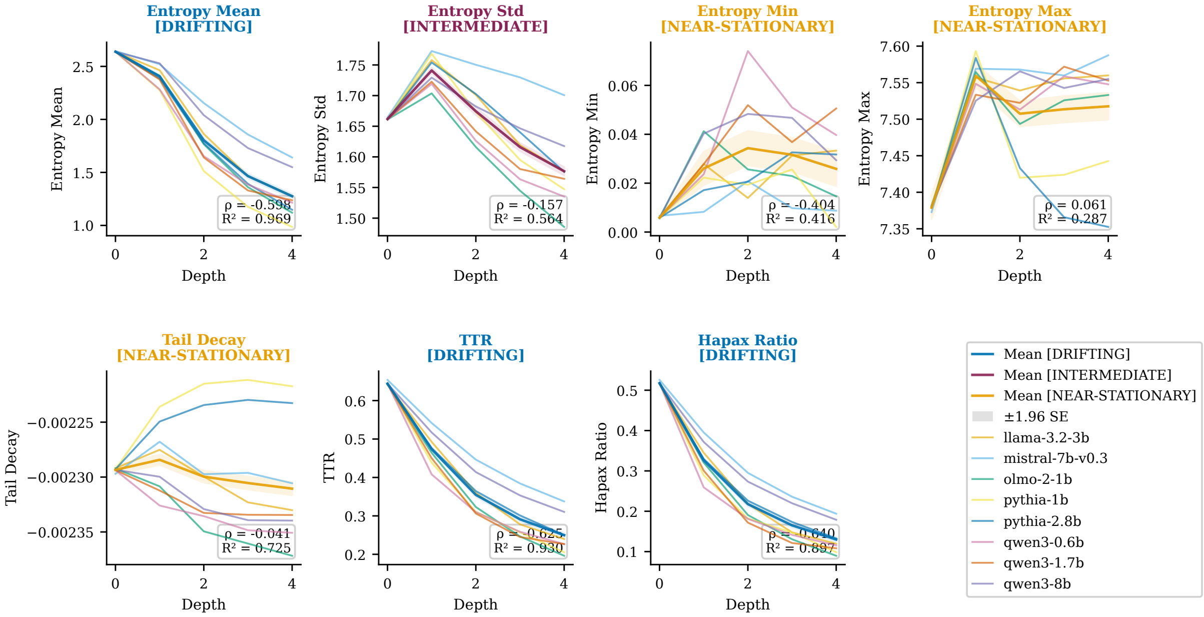

Figure 8: Per-feature depth trends. Bold line: pooled mean (±1.96 SE shading) across all generators, colored by DDR zone (blue: drifting, wine: intermediate, amber: near-stationary). Thin traces: individual generators. Spearman ρ and linear R^2 are annotated per panel.

q Cross-Temperature Transfer

Figure 10 shows cross-temperature transfer results for the 6-generator LOGO pipeline. We evaluate four train→test temperature conditions using Qwen3-1.7B data regenerated at t=1.0 (remaining five generators retain t=0.7).[6]

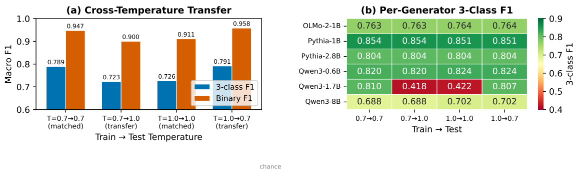

Figure 10: Cross-temperature transfer evaluation (6-gen LOGO). (a) Overall F1 by train→test temperature condition. Matched conditions (0.7→0.7, 1.0→1.0) outperform cross-temperature transfer, with 0.7→1.0 showing the largest degradation (−6.6 pp in 3-class F1). (b) Per-generator 3-class F1 heatmap: Qwen3-1.7B shows dramatic degradation under cross-temperature conditions (0.418) while other generators remain stable. The temperature mismatch disproportionately affects the generator whose data was regenerated.

Key observations from the cross-temperature analysis:

1. **Matched-temperature robustness**: Both t=0.7 and t=1.0 matched conditions yield comparable performance (3-class F1: 0.789 vs. 0.726), with the gap attributable to weaker depth signal at high temperature rather than classifier degradation.
2. **Asymmetric transfer**: Training at t=1.0 and testing at t=0.7 (F1 = 0.791) transfers well, but the reverse (0.7→1.0, F1 = 0.723) degrades. This asymmetry reflects the information structure: features learned from high-variance (high-temperature) data generalize to low-variance settings, but not vice versa.
3. **Generator-specific vulnerability**: Qwen3-1.7B, the only generator with temperature-mismatched test data, drops to F1 = 0.418 under cross-temperature conditions while all other generators maintain F1 $>$ 0.76. This confirms that depth features are calibrated to the specific decoding regime.
4. **Binary transfer for the mismatched generator**: For Qwen3-1.7B (the only generator with actual temperature mismatch), binary F1 drops from 0.948 to 0.713 under 0.7→1.0 transfer but recovers in the reverse direction (1.0→0.7: 0.948), consistent with the asymmetric pattern in Item 2. The aggregate binary range (0.900–0.958) is inflated by five generators whose data is identical across conditions.

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[6] The t=1.0 data uses top-p=0.95 (vs. p=0.9 for other configurations); this difference may contribute to the observed degradation beyond the temperature effect alone.

Feature set	**F1**	**Top feature (imp.)**
Full (7 features)	0.773	hapax_ratio (0.35)
Surface only (2)	0.688	hapax_ratio (0.41)
Reference only (5)	0.713	entropy_mean (0.27)

Table 24: Feature ablation (8-gen, 10-fold stratified CV, 3-class). Full > surface (+8.5 pp) and Full > reference (+6.0 pp) confirms complementarity.

r Feature Ablation

This appendix presents feature ablation results supporting the complementarity analysis in §4.2.

s Feature Correlation Matrix

This appendix documents inter-feature correlations and the PCA analysis referenced in §4.2.

TTR and hapax ratio are near-perfectly correlated ($r = 0.97$), and both correlate strongly with mean entropy ($r = 0.86$ and 0.81). These three drifting features form a tightly coupled cluster: PCA on the standardized drifting-feature subset shows that a single principal component explains 92.3% of their joint variance, with near-uniform loadings (0.56–0.59). These three features thus share a single latent dimension. Full 7-feature PCA yields PC1 (44.4% variance) loading on the drifting triplet (0.53–0.55) and PC2 (18.3% variance) capturing entropy distributional shape (entropy std vs. entropy min contrast). In contrast, $H_{\max}$, τ (tail decay), and $H_{\min}$ show weak pairwise correlations with all other features ($|r| < 0.27$); these features capture independent aspects of the token-probability distribution.

Table 25: Pearson correlation matrix of the seven depth features ($N = 27{,}529$).

	$\bar{H}$	σ_H	$H_{\min}$	$H_{\max}$	τ	TTR	Hapax
$\bar{H}$	—	0.49	0.31	0.01	0.07	**0.86**	**0.81**
σ_H	0.49	—	−0.15	0.16	0.26	0.43	0.34
$H_{\min}$	0.31	−0.15	—	−0.04	0.01	0.20	0.23
$H_{\max}$	0.01	0.16	−0.04	—	−0.08	−0.13	−0.15
τ	0.07	0.26	0.01	−0.08	—	0.13	0.14
TTR	**0.86**	0.43	0.20	−0.13	0.13	—	**0.97**
Hapax	**0.81**	0.34	0.23	−0.15	0.14	**0.97**	—

t Cross-Family Reference Models

u Independent-Seed Control

To test for topic-level leakage from shared seeds across generators, we construct an independent-seed evaluation where Pythia-2.8B and OLMo-2-1B use completely disjoint Wikipedia seed pools (500 seeds each, zero overlap). Each generator produces depth-0 through depth-4 chains on its own seed pool; features are extracted with the same Pythia-410M reference model. We then train a 7-feature XGBoost classifier on one generator's data and test on the other, in both directions.

The small gap (-1.4 pp) between independent-seed and shared-seed evaluation falls within bootstrap confidence intervals, and both directions individually exceed 0.66. Combined with the seed-level demeaning control (§4.1), these results provide converging evidence that depth classification is driven by distributional properties of recursive generation, not topic identity.

Semantic similarity analysis. Embedding cosine similarity between consecutive depth steps increases monotonically: 0→1 (0.622), 1→2 (0.683), 2→3 (0.801), 3→4 (0.859). Deeper generations are thus *more* semantically similar to their inputs, yet the classifier still distinguishes depth classes, suggesting that the depth signal resides in distributional rather than topical properties. Prompt type modulates the baseline similarity level (continuation $0.717 <$ elaboration $0.809 \approx$ rewriting 0.808), plausibly because semantic-preservation constraints reduce topic drift while leaving the distributional location shift intact.

Reference	**Family**	**Params**	**LOGO F1**	**Ent. Drift**
Pythia-410M	Pythia	410M	0.803	$[-1.19, -1.40]$
OLMo-2-1B	OLMo	1.5B	0.807	$[-0.61, -0.91]$
Qwen3-0.6B	Qwen	0.6B	0.816	$[-0.96, -1.35]$

Table 26: Cross-family reference model validation (5-gen LOGO subset). Drift direction is consistent (negative) across all three references; magnitude varies with capacity. Spread in LOGO F1 is 0.013.

Setting	**Binary F1**	**95% CI**
Pythia-2.8B → OLMo-2-1B	0.695	[0.673, 0.717]
OLMo-2-1B → Pythia-2.8B	0.666	[0.643, 0.689]
Independent-seed average	0.680	—
Shared-seed LOGO (reference)	0.694	[0.688, 0.700]

Table 27: Independent-seed cross-generator evaluation (binary: gen-1 vs. gen-2+). Disjoint seed pools eliminate any possibility of topic-based leakage. The average independent-seed F1 (0.680) is within 1.4 pp of the shared-seed LOGO baseline (0.694). The classifier thus captures generative degradation patterns rather than seed-specific topic content.

v Per-Generator Equivalence Testing

To confirm that the near-stationary classification is not a pooling artifact, we apply two one-sided t-tests (TOST) (Lakens, 2017) for each feature within each generator separately (gen-1 vs. gen-2+, equivalence bound $|d| < 0.20$). We adopt Cohen's conventional small effect size (d=0.20) (Cohen, 1988) as the smallest effect of practical significance (SESOI): effects below this threshold are unlikely to be practically meaningful for depth discrimination. LOFO analysis independently supports this choice: removing any near-stationary feature changes classification F1 by ≤1.5 pp (Appendix f), linking the statistical threshold to classification irrelevance. Sensitivity analysis at $\varepsilon \in \{0.15, 0.20, 0.25\}$ confirms the partition is robust to the bound choice (Appendix w); bootstrap resampling (B=1,000) shows 100% rejection stability for all near-stationary features (Appendix v.1); and the partition is invariant across $\varepsilon \in [0.10, 0.75]$ (Appendix w). Table 28 reports observed $|d|$; ✓ marks features for which the TOST null hypothesis (non-equivalence) is rejected at $\alpha = 0.05$.

Entropy min passes equivalence for all eight generators; entropy max passes for six of eight (exceeding ε only for Pythia-1B and Pythia-2.8B, where $|d| \approx 0.21$–0.25); tail decay passes for four of eight. All drifting and intermediate features fail equivalence uniformly, with $|d|$ ranging from 0.23 (entropy std, Mistral) to 1.40 (entropy mean, Pythia-1B). The partition is robust at the individual-generator level, with entropy min as the most consistently stationary feature. Figure 11 visualizes the DDR partition in ($|\rho|$, d) space, showing a clear gap between the three groups.

Table 28: Per-generator TOST equivalence (ε=$|d|$<0.20, gen-1 vs. gen-2+). Observed $|d|$; ✓ = equivalent ($p_{\mathrm{TOST}} < 0.05$). Features grouped by DDR partition.

	Drifting			**Int.**	**Near-stationary**		
Generator	$\bar{H}$	TTR	hapax	σ_H	$H_{\max}$	tail	$H_{\min}$
Pythia-2.8B	1.20	0.77	0.67	0.39	0.25	0.07 ✓	0.03 ✓
Pythia-1B	1.40	0.80	0.65	0.53	0.21	0.08 ✓	0.02 ✓
Qwen3-1.7B	1.12	0.93	0.83	0.52	0.02 ✓	0.09 ✓	0.05 ✓
Llama-3.2-3B	1.15	1.04	0.98	0.50	0.01 ✓	0.19	0.01 ✓
OLMo-2-1B	1.19	1.08	1.00	0.61	0.06 ✓	0.22	0.07 ✓
Qwen3-8B	0.86	0.71	0.66	0.37	0.04 ✓	0.15	0.01 ✓
Qwen3-0.6B	0.92	0.63	0.50	0.56	0.01 ✓	0.08 ✓	0.10 ✓
Mistral-7B	0.89	0.82	0.82	0.23	0.00 ✓	0.16	0.04 ✓
Equiv. count	0/8	0/8	0/8	0/8	6/8	4/8	8/8

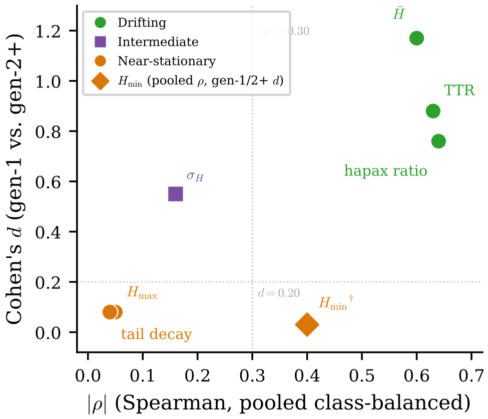

Figure 11: DDR partition in pooled ($|\rho|$, Cohen's d) space. Drifting features (green) cluster at high $|\rho|$ and d; near-stationary features (orange) cluster near the origin; entropy std (purple) occupies a sole intermediate position. Dashed lines: TOST equivalence bound (d=0.20) and $|\rho|$=0.30. Diamond marker: entropy min (pooled $|\rho|$=0.40 reflects the gen-0/gen-1 boundary; gen-1 vs. gen-2+ d=0.03).

v.1 TOST Bootstrap Stability

To verify that the equivalence conclusions in §v are not sensitive to sample composition, we assess their stability under bootstrap resampling. For each of four representative features, we run 1,000 bootstrap iterations (base seed = 42): each iteration independently resamples with replacement from the gen-1 and gen-2+ pools, recomputes Cohen's d and the TOST p-value (equivalence margin $|d| < 0.20$).

All three near-stationary features achieve a 100% rejection rate across 1,000 bootstrap iterations, and their effect-size 95% CIs fall entirely within the

Table 29: Bootstrap stability of TOST equivalence (B=1,000, margin $|d|$<0.20). Rejection rate = fraction of bootstrap iterations where TOST rejects non-equivalence at α=0.05.

Feature	Rejection rate	Median d	d 95% CI
entropy_max	100% (1000/1000)	0.062	[0.037, 0.088]
entropy_min	100% (1000/1000)	−0.016	[−0.042, 0.011]
tail_decay	100% (1000/1000)	0.088	[0.065, 0.111]
entropy_std	0% (0/1000)	0.458	[0.435, 0.479]

±0.20 equivalence margin. Entropy std, included as an intermediate control, is never rejected (0%), with its CI $[0.435, 0.479]$ lying entirely outside the margin. No feature exhibits boundary-crossing: the classification of each feature is unanimous across all bootstrap samples, confirming that the equivalence conclusions are robust to resampling variability.

w DDR Partition Validation

Data-Driven Clustering

To assess the DDR partition structure and determine whether the drifting/non-drifting split is data-driven, we apply k-means clustering ($k \in \{2, 3, 4, 5\}$) to the seven features in standardized $(|\rho|, d)$ space (Table 30).

Table 30: k-means clustering of 7 features in $(|\rho|, d)$ space.

k	Silhouette	Clusters
2	**0.660**	$\{\bar{H}$, TTR, hapax$\}$ | $\{\sigma_H, H_{\min}, H_{\max}$, tail$\}$
3	0.532	$\{\bar{H}$, TTR, hapax$\}$ | $\{\sigma_H$, $H_{\max}$, tail$\}$ | $\{H_{\min}\}$
4	0.527	$\{\bar{H}$, TTR, hapax$\}$ | $\{H_{\max}$, tail$\}$ | $\{\sigma_H\}$ | $\{H_{\min}\}$

The primary drifting/non-drifting split (k=2, silhouette = 0.660) is the strongest cluster structure. At k=3, entropy min forms its own cluster rather than joining the near-stationary group because its pooled $|\rho|$=0.40 (driven by the gen-0/gen-1 boundary) places it far from other near-stationary features in ρ space. The 3/1/3 DDR partition resolves this by using the gen-1 vs. gen-2+ Cohen's d (d=0.03) to place entropy min in the near-stationary group, a domain-informed refinement that unsupervised clustering cannot recover from pooled statistics alone. At k=4 (silhouette = 0.571), entropy std separates from the near-stationary group, consistent with its intermediate DDR classification.

Exhaustive partition enumeration. To confirm that the three-vs-four split is optimal rather than incidental, we enumerate all $\binom{7}{3}$=35 possible ways to assign three features to a "drifting" group and the remaining four to "non-drifting," computing the silhouette score for each partition in standardized $(|\rho|, d)$ space. The observed DDR partition ({entropy mean, TTR, hapax ratio}) achieves the highest silhouette (0.660), with a gap of 0.18 to the second-ranked split; 20 of 35 partitions produce negative silhouette scores.

Per-generator replication. To rule out the possibility that the pooled silhouette is an artifact of mixing heterogeneous generators, we repeat the $(|\rho|, d)$ clustering independently for each of the eight generators. Every generator recovers the identical $\{\bar{H}$, TTR, hapax$\}$ vs. $\{\sigma_H$, $H_{\min}$, $H_{\max}$, tail$\}$ partition, with silhouette scores ranging from 0.505 (Pythia-1B) to 0.760 (Mistral-7B-v0.3; mean = 0.651). No generator falls below the 0.50 moderate-structure threshold, confirming that the two-cluster structure is a per-generator property rather than a pooling artifact.

TOST Equivalence Bound Sensitivity

Table 31 shows the fraction of generators for which each feature passes the TOST equivalence test (gen-1 vs. gen-2+, α=0.05) at three equivalence bounds ε.

Table 31: TOST equivalence pass rate across 8 generators at varying ε.

DDR Zone	Feature	ε=0.15	ε=0.20	ε=0.25
Drifting	$\bar{H}$ (ent. mean)	0/8	0/8	0/8
	TTR	0/8	0/8	0/8
	hapax ratio	0/8	0/8	0/8
Intermediate	σ_H (ent. std)	0/8	0/8	0/8
Near-stat.	$H_{\min}$ (ent. min)	7/8	8/8	8/8
	$H_{\max}$ (ent. max)	6/8	6/8	6/8
	tail decay	4/8	4/8	7/8

The partition is robust to the equivalence bound: drifting features reject equivalence at all ε for all generators, while entropy min passes even at the stricter bound ε=0.15 (7/8 generators). Tail decay is the most ε-sensitive near-stationary feature, passing 4/8 at ε=0.20 but 7/8 at ε=0.25; the four non-passing generators show $|d|$=0.15–0.22, still well below the drifting threshold ($d > 0.50$). Entropy max is stable across all three bounds (6/8), with the two failures (Pythia-1B d=0.21, Pythia-2.8B d=0.25) reflecting same-family reference affinity with the Pythia-410M reference model.

Threshold sensitivity. To address concerns about the sensitivity of the drifting-boundary threshold, we sweep $|d|$ from 0.10 to 0.80 in increments of 0.05 and record the resulting binary partition (drifting vs. stationary) along with the silhouette score in standardized $(|\rho|, d)$ space. Across the entire range, only *three* distinct partitions emerge, reflecting natural gaps between feature clusters rather than threshold artifacts. The three-feature drifting set (entropy mean, TTR, hapax ratio) is stable across $|d| \in [0.50,\ 0.75]$ with a silhouette of 0.660—the peak value in the sweep. Including the intermediate feature entropy std (d=0.49) yields a 4/3 partition stable across $|d| \in [0.10,\ 0.45]$ (silhouette = 0.451). The partition used in our analysis corresponds to the peak-silhouette region, confirming that the drifting/stationary boundary reflects a natural gap in the $(|\rho|, d)$ landscape rather than an artifact of a particular threshold choice.

Partition stability across ε thresholds. The DDR partition classifies features as non-stationary ($d \geq \varepsilon$) or stationary ($d < \varepsilon$) based on pooled Cohen's d (§4.2). Table 32 reports the critical ε at which each feature's classification changes; the three drifting and three near-stationary features maintain their classification for all $\varepsilon \in [0.10, 0.75]$.

Table 32: Partition stability under varying ε. Critical threshold: the ε above which a feature is reclassified as stationary. The drifting/near-stationary split is invariant for $\varepsilon \in [0.10, 0.75]$; only entropy std ($d$=0.55) changes at $\varepsilon > 0.55$.

DDR Zone	**Feature**	**Pooled d**	**Stationary at $\varepsilon >$**
Drifting	$\bar{H}$ (ent. mean)	1.17	1.17
	TTR	0.88	0.88
	hapax ratio	0.76	0.76
Intermediate	σ_H (ent. std)	0.55	0.55
Near-stat.	$H_{\max}$ (ent. max)	0.08	0.08
	tail decay	0.08	0.08
	$H_{\min}$ (ent. min)	0.016	0.016

The 3/1/3 partition is stable across $\varepsilon \in [0.10, 0.75]$: the three drifting features remain non-stationary at all $\varepsilon < 0.76$, the three near-stationary features remain stationary at all $\varepsilon > 0.08$, and only entropy std changes classification (becoming stationary at $\varepsilon > 0.55$), confirming it as the sole threshold-sensitive feature. The minimum gap between adjacent zone boundaries ($d_{\text{hapax}} - d_{\sigma_H} = 0.21$) exceeds typical effect-size estimation uncertainty, providing a margin against sampling noise.

Holdout Generator Split Validation

To confirm the DDR partition is not an artifact of the specific generator pool, we split the eight generators into disjoint selection (4) and validation (4) sets, compute per-subset pooled $(|\rho|, d)$, and check whether all seven features receive the same zone assignment in both subsets. We evaluate two complementary splits to avoid dependence on a single partition: Split A selects {Pythia-2.8B, Qwen3-1.7B, OLMo, Mistral} (validation: Pythia-1B, Qwen3-0.6B, Qwen3-8B, Llama); Split B selects {Pythia-1B, Qwen3-0.6B, OLMo, Mistral} (validation: Pythia-2.8B, Qwen3-1.7B, Qwen3-8B, Llama).

Table 33: Holdout split validation of the DDR partition. Selection (sel) and validation (val) $|\rho|$ and d for two disjoint 4/4 generator splits. Zone assignment is identical across all four subsets for every feature.

		Split A			**Split B**		
Zone	**Feature**	sel $\lvert\rho\rvert$	val $\lvert\rho\rvert$	$\Delta\lvert\rho\rvert$	sel $\lvert\rho\rvert$	val $\lvert\rho\rvert$	$\Delta\lvert\rho\rvert$
Drifting	$\bar{H}$ (ent. mean)	0.60	0.58	0.02	0.60	0.58	0.02
	TTR	0.61	0.59	0.02	0.60	0.59	0.01
	hapax ratio	0.62	0.60	0.02	0.62	0.61	0.01
Intermediate	σ_H (ent. std)	0.19	0.21	0.02	0.22	0.17	0.05
Near-stat.	$H_{\min}$ (ent. min)	0.39	0.36	0.03	0.38	0.37	0.01
	$H_{\max}$ (ent. max)	0.05	0.05	0.00	0.04	0.06	0.02
	tail decay	0.10	0.08	0.02	0.11	0.07	0.04

Under simple joint $(|\rho|, d)$ thresholds, both splits produce a 3/2/2 partition: entropy min is classified as a second intermediate feature due to its elevated pooled $|\rho|$ (0.36–0.39). However, entropy min's binary (gen-1 vs. gen-2+) d values are uniformly below 0.06 across all four split×subset combinations, consistent with near-stationarity on the task-relevant metric; its pooled $|\rho|$ reflects change concentrated at the gen-0/gen-1 boundary rather than sustained drift. Five of seven features are unambiguously classified under any reasonable criterion; entropy min's boundary status is resolved by the binary-task d criterion.

Selection and validation subsets produce identical partitions in both splits (Table 33; Figure 12). The maximum $\Delta|\rho|$ between selection and validation is 0.05 (entropy std, Split B), and the minimum gap between adjacent zones remains >0.15 in all subsets.

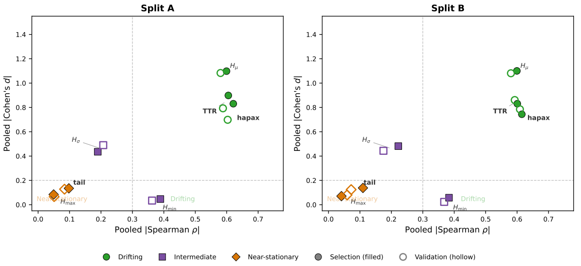

Figure 12: Holdout split validation in $(|\rho|, d)$ space. Filled markers: selection set; hollow: validation set. Zone assignments are identical across subsets; selection/validation pairs cluster tightly for all features.

Heteroscedasticity Sensitivity

Levene tests reject equal variance between gen-1 and gen-2+ for drifting features ($p < 10^{-10}$; variance ratios 1.86–2.21). To verify that the DDR partition is robust to this heteroscedasticity, we supplement pooled Cohen's d with Glass's Δ (using the gen-1 standard deviation as denominator) and Cliff's δ (a non-parametric ordinal effect size; Table 34).

Table 34: Effect size comparison (gen-1 vs. gen-2+, $N = 29{,}095$). Glass's Δ uses the gen-1 SD as denominator; Cliff's δ thresholds follow Romano et al. (2006): $|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium, otherwise large.

Zone	**Feature**	$\lvert d\rvert$	$\lvert\Delta\rvert$	$\lvert\delta\rvert$
Drifting	ent. mean	1.054	1.410	0.587
	TTR	0.814	1.116	0.510
	hapax ratio	0.735	0.936	0.528
Intermediate	ent. std	0.457	0.605	0.295
Near-stat.	tail decay	0.087	0.099	0.072
	ent. max	0.062	0.065	0.021
	ent. min	0.016	0.015	$0.354^{\dagger}$

Effect sizes in Table 34 are computed on the binary task (gen-1 vs. gen-2+); the main text (§4.2) reports 3-class pooled d values (e.g., hapax ratio d=0.76), which are slightly larger because the gen-0/gen-1 contrast contributes additional separation. Glass's Δ amplifies drifting effects (1.41 vs. 1.05 for entropy mean) because gen-2+ variance exceeds gen-1 variance (ratio 2.08), making the pooled Cohen's d conservative. Near-stationary features remain negligible under both parametric metrics ($|\Delta| \leq 0.099$). The minimum drifting-to-maximum-near-stationary gap is 9.5× under Glass's Δ (0.936 / 0.099); heteroscedasticity does not erode the partition. Cliff's δ classifies all drifting features as "large" ($|\delta| > 0.474$) and entropy std as "small" ($|\delta|$=0.295), consistent with the parametric partition.

†Entropy min's elevated Cliff's δ (0.354, "medium") reflects distributional shape asymmetry between gen-1 and gen-2+ that rank-based statistics detect but location measures do not (Cohen's d=0.016, Glass's Δ=0.015). This is consistent with its documented boundary status (§4.2): the gen-0/gen-1 transition produces rank-order shifts that persist into deeper generations without affecting the mean.

x Generator Subset Summary

Table 35 summarizes the generator subsets used across experiments.

Table 35: Generator subsets by experiment type.

Experiment	N	**Generators**
Main LOGO (Table 2)	8	All: Py-2.8B/1B, OLMo, Qw-1.7B/0.6B/8B, Llama, Mistral
Cross-model chains (Table 26)	5	Py-2.8B/1B, OLMo, Qw-1.7B/0.6B
QLoRA (Table 7)	6	Py-2.8B/1B, OLMo, Qw-1.7B/0.6B/8B
Temperature ablation (Table 12)	4	Py-2.8B, OLMo, Qw-1.7B/0.6B
Prompt-type (Appendix c)	1	Py-2.8B only
Cross-domain (Table 16)	3	Py-1B, Qw-1.7B, Mistral
Instruct (Table 17)	3	OLMo, Qw-1.7B, Llama
Downstream (Appendix n)	5	Py-2.8B/1B, OLMo, Qw-1.7B/0.6B

The 5-generator set was the initial development subset; Qwen3-8B was added for scale evaluation, and Llama-3.2-3B and Mistral-7B-v0.3 were added last for family-diversity testing. Subset differences reflect chronological expansion rather than selective reporting.

y Qualitative Examples

Table 36 presents an example generation chain from seed 120 (Pythia-2.8B, continuation prompt), illustrating how text evolves across recursive generation steps. Table 37 reports the corresponding 7-feature values. Central-tendency features ($\bar{H}$, TTR, hapax ratio) drift monotonically while shape features ($H_{\max}$, tail decay) remain near-stationary, consistent with the differential depth response pattern.

z Cross-Generator Transfer Analysis

To isolate the depth signal from generator identity, we train a binary classifier (gen-1 vs. gen-2+) on each of the eight generators independently and evaluate on all others. Table 38 reports the full 8×8 transfer matrix and Figure 13 visualizes it as a heatmap. Note that these classifiers are trained with a binary objective, whereas the LOGO classifiers in Table 2 are trained on a 3-class task and

Table 36: Example text chain from seed 120 (Pythia-2.8B, continuation prompt). Gen-0 is the original Wikipedia passage; gen-1 and gen-2 are successive LLM continuations. Text is truncated for space.

Gen-0 (Wikipedia seed)
Aquarius is an equatorial constellation of the zodiac, between Capricornus and Pisces. Its name is Latin for “water-carrier” or “cup-carrier”. Aquarius is one of the oldest of the recognized constellations along the zodiac (the Sun’s apparent path). It was one of the 48 constellations listed by the 2nd century astronomer Ptolemy, and it remains one of the 88 modern constellations. It is found in a region often called the Sea due to its profusion of constellations with watery associations such as Cetus the whale, Pisces the fish, and Eridanus the river. [. . .]
Gen-1
Babylonian astrology and was also depicted in ancient Egyptian art. In Greek mythology, Aquarius is the son of Poseidon and Gaia. He is also the god of the sea and the sky, the god of travelers, and the guardian of the underworld. In Roman mythology, Aquarius is the son of Saturn and the nymph Egeria, and the god of the sea. In Jewish mythology, Aquarius is the son of the God of the Air and the Goddess of the Moon. In Chinese mythology, Aquarius is the son of the God of the Sky and the Goddess of the Earth. [. . .]
Gen-2
God of the Sky and the Goddess of the Earth. In the ancient Greek mythology, Aquarius is the son of the God of the Sky and the Goddess of the Earth. In the Hindu mythology, Aquarius is the son of the God of the Sky and the Goddess of the Earth. In the ancient Egyptian mythology, Aquarius is the son of the God of the Sky and the Goddess of the Earth. [. . .]

evaluated on the binary subtask; the two setups are therefore not directly comparable.

All 56 off-diagonal transfer pairs yield F1 above chance, with a mean of 0.810 (95% bootstrap CI [0.796, 0.822]; one-sample $t = 46.58$, $p < 10^{-10}$). Transfer is strongest within families (e.g., Py-2.8B→Py-1B: 0.87) and weakest for Py-1B→Mi-7B (0.62), consistent with the scale and reference-affinity effects discussed in §4.2. These results confirm that depth signatures generalize across architectures, parameter scales, and model families, providing evidence against generator identity as a primary confound.

Table 37: Feature values for the chain in Table 36. Drifting features ($\bar{H}$, TTR, hapax) show monotonic degradation; shape features ($H_{\max}$, tail decay) remain stable (range <2% and <4% respectively).

Feature	Gen-0	Gen-1	Gen-2
$\bar{H}$ (entropy mean)	2.720	2.206	0.780
σ_H (entropy std)	1.772	1.816	1.517
$H_{\min}$ (entropy min)	0.002	0.005	0.000
$H_{\max}$ (entropy max)	7.000	7.016	6.926
tail decay	−0.00227	−0.00231	−0.00223
TTR	0.659	0.272	0.081
hapax ratio	0.561	0.183	0.005

Train \ Test	**Py-2.8B**	**Py-1B**	**Qw-0.6B**	**Qw-1.7B**	**Qw-8B**	**OLMo-1B**	**Ll-3B**	**Mi-7B**
Py-2.8B	0.985	0.873	0.823	0.833	0.741	0.829	0.815	0.712
Py-1B	0.834	0.987	0.828	0.821	0.704	0.778	0.764	0.620
Qw-0.6B	0.786	0.846	0.991	0.851	0.786	0.795	0.760	0.680
Qw-1.7B	0.794	0.839	0.859	0.989	0.790	0.817	0.798	0.717
Qw-8B	0.811	0.836	0.853	0.855	0.984	0.848	0.839	0.790
OLMo-1B	0.811	0.827	0.837	0.852	0.803	0.988	0.828	0.763
Ll-3B	0.856	0.860	0.845	0.851	0.798	0.851	0.987	0.787
Mi-7B	0.834	0.837	0.837	0.845	0.817	0.849	0.851	0.969

Table 38: Cross-generator binary F1 transfer matrix (gen-1 vs. gen-2+). Each row trains on one generator; each column evaluates on another. Diagonal entries (underlined) are within-generator. All 56 off-diagonal pairs exceed chance (0.5). Classifiers use the same 7-feature XGBoost configuration as LOGO but are trained with a binary objective.

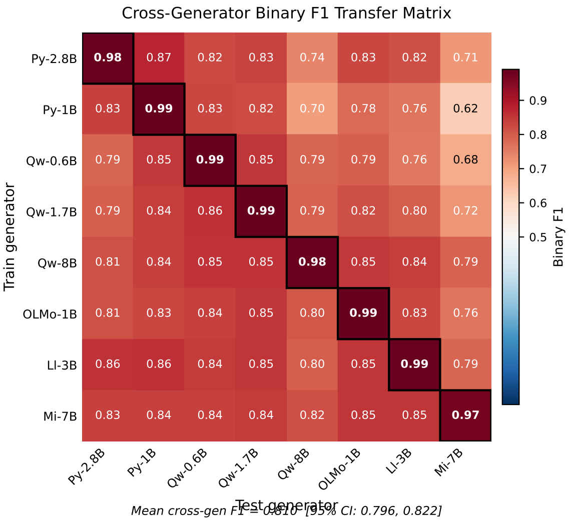

Figure 13: Heatmap of the cross-generator binary F1 transfer matrix (Table 38). Same-family pairs (e.g., Pythia↔Pythia, Qwen↔Qwen) show slightly higher transfer; Mi-7B is the hardest transfer target (column minimum 0.62) yet remains well above chance.

aa Supplementary DDR Characterization

aa.1 Partition Stability Across Temperatures

To assess whether the DDR three-way partition generalizes beyond the default temperature (τ=0.7), we computed per-feature Cohen’s d (generation 1 vs. generation 2+) at $\tau \in \{0.5, 0.7, 0.9\}$ for all four generators in the temperature ablation (12 conditions total; Table 39).

Table 39: Mean $|d|$ (gen-1 vs. gen-2+) across 4 generators at each temperature. Partition labels: **D**rifting ($|d| > 0.5$), **I**ntermediate ($0.1 < |d| < 0.5$), near-**S**tationary ($|d| < 0.1$). Boldface marks features that shift from their τ=0.7 partition.

Feature	**τ=0.5**	**τ=0.7**	**τ=0.9**	**Group**
Entropy mean	1.39 ^{D}	1.10 ^{D}	0.58 ^{D}	Drifting
TTR	1.00 ^{D}	0.85 ^{D}	**0.45** ^{I}	Drifting
Hapax ratio	0.87 ^{D}	0.75 ^{D}	**0.42** ^{I}	Drifting
Entropy std	**0.84** ^{D}	0.52 ^{D}	**0.20** ^{I}	Intermediate
Entropy max	0.06 ^{S}	0.09 ^{S}	**0.11** ^{I}	Stationary
Tail decay	0.07 ^{S}	**0.12** ^{I}	**0.15** ^{I}	Stationary
Entropy min	0.04 ^{S}	0.06 ^{S}	0.04 ^{S}	Stationary

The partition anchors are robust: entropy mean remains drifting in all 12 conditions, TTR in 10/12, and hapax ratio in 9/12; entropy min and entropy max stay near-stationary in 10/12 and 9/12 conditions, respectively. The two boundary features are more temperature-sensitive: entropy std is drifting at τ=0.5 for all generators but falls to intermediate at τ=0.9 for three of four, while tail decay fluctuates between near-stationary and intermediate across generators. Effect sizes compress monotonically with temperature: for the fixed drifting set (entropy mean, TTR, hapax ratio), mean $|d|$ drops from 1.09 at τ=0.5 to 0.90 at τ=0.7 to 0.48 at τ=0.9, shifting individual features across partition boundaries at high temperature without altering the qualitative hierarchy: lexical diversity features drift fastest, entropy extrema remain stable across all conditions tested.

aa.2 Per-Feature Convergence Dynamics

This appendix visualizes per-feature convergence trajectories across generation depth, complementing the DDR analysis in §4.2.

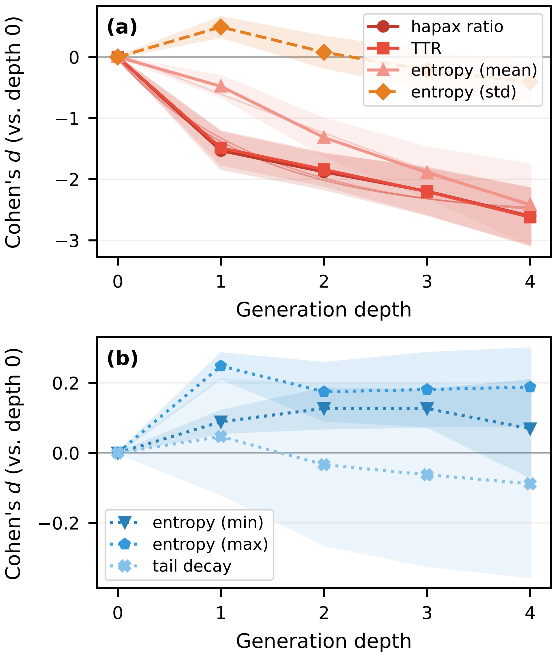

Figure 14: Per-feature convergence dynamics across generation depths 0–4, averaged over 8 generators. Each feature is expressed as Cohen's d relative to depth 0, so all features start at zero and the y-axis directly reflects effect size. (a) Drifting features (hapax ratio, TTR, entropy mean) and the intermediate feature (entropy std) diverge monotonically; lexical diversity features (TTR, hapax ratio) form a fast-convergence group ($|d| > 2$ by depth 2), while entropy mean and entropy std drift more slowly. (b) Near-stationary features (entropy max, tail decay, entropy min) remain within $|d| < 0.25$ across all depths, confirming location stability. Shaded bands indicate ± 1 standard deviation across generators.

aa.3 DDR Causality Analysis

This appendix validates the causal predictive power of DDR features through three complementary analyses: permutation importance, Granger causality, and feature intervention.

Permutation Importance. Table 40 reports the mean F1 drop when each feature's values are randomly shuffled across the test set (baseline F1 = 0.771). Drifting features dominate: their average F1 drop (0.187) is $5.5\times$ larger than the near-stationary average (0.034), confirming that location-shift features, not shape features, drive depth classification.

Granger Causality. Table 41 reports Fisher-combined p-values from panel Granger causality tests (lag = 1) across 8 generators (N = 29,095). Six of seven features exhibit significant Granger causality ($p < 0.001$), meaning past feature val-

Feature	Group	Mean F1 Drop	Rank
hapax_ratio	Drifting	0.235	1
entropy_mean	Drifting	0.170	2
ttr	Drifting	0.156	3
entropy_max	Near-stat.	0.070	4
entropy_std	Intermediate	0.057	5
entropy_min	Near-stat.	0.019	6
tail_decay	Near-stat.	0.014	7

Table 40: Permutation importance (8-gen LOGO). Each feature is shuffled independently; the resulting F1 drop quantifies marginal importance. Drifting features average 0.187 vs. near-stationary 0.034 ($5.5\times$).

ues predict future depth beyond what past depth alone explains. The sole exception is entropy_min ($p = 0.375$), providing additional evidence for its near-stationary classification: it carries neither large marginal importance (Table 40, rank 7) nor temporal predictive direction.

Feature	Combined p	ΔR^2	Significant
entropy_std	< 0.001	0.031	Yes
hapax_ratio	< 0.001	—	Yes
ttr	< 0.001	—	Yes
entropy_mean	< 0.001	—	Yes
entropy_max	< 0.001	—	Yes
tail_decay	< 0.001	—	Yes
entropy_min	0.375	—	No

Table 41: Granger causality (Fisher combined across 8 generators, lag = 1, $N = 29{,}095$). ΔR^2 is reported where available.

Key Findings.

1. **Drifting dominance.** The $5.5\times$ permutation-importance gap confirms that drifting features (entropy_mean, ttr, hapax_ratio) are the primary depth signal carriers, consistent with the DDR partition derived from convergence analysis (§4.2).
2. **Entropy_min lacks causal direction.** Beyond its small effect size (F1 drop = 0.019), entropy_min is the only feature without significant Granger causality ($p = 0.375$). This strengthens the near-stationary classification: the feature is neither marginally important nor temporally predictive.
3. **Entropy_max asymmetry.** Entropy_max ranks fourth in permutation importance (F1 drop = 0.070) but produces the largest degradation in the counterfactual intervention (-2.0 pp), indicating that it provides unique signal not redundant with other features; its causal contribution exceeds its marginal importance.
4. **Mistral heterogeneity.** In per-generator Granger tests, Mistral-7B shows non-significant hapax_ratio and ttr ($p > 0.4$), though Fisher combination across generators remains significant. This generator-specific attenuation partly explains Mistral's lowest LOGO F1 (0.605) and suggests that DDR fingerprints are weaker in models with heavier repetition penalties.

aa.4 Decoding Strategy Robustness

To verify that DDR shape stability is not an artifact of nucleus sampling (p=0.9, τ=0.7), we repeat the generation chain experiment under two alternative decoding strategies: greedy decoding (τ=0) and top-k sampling (k=50, τ=0.7). All eight generators spanning five architectural families produce ~1,000 chains of depth 0–4 per strategy, yielding ~5,000 texts per generator–strategy combination.

Table 42 reports Cohen's d (gen-1 vs. gen-2+) for the three shape features across strategies; Table 43 provides the corresponding values for drifting and intermediate features.

Table 42: Shape-feature Cohen's d (gen-1 vs. gen-2+) under three decoding strategies per generator. Values below 0.20 indicate negligible drift. Shape features are expected to remain near-stationary across strategies if DDR is decoding-agnostic.

		entropy_max			entropy_min			tail_decay		
Family	Generator	Nuc.	Gr.	Top-k	Nuc.	Gr.	Top-k	Nuc.	Gr.	Top-k
Pythia	2.8B	0.25	0.09	0.26	0.03	0.08	0.04	0.07	0.07	0.12
	1B	0.21	0.12	0.27	0.02	0.04	0.04	0.08	0.04	0.18
OLMo	2-1B	0.06	0.03	0.01	0.07	0.03	0.01	0.22	0.03	0.22
Qwen3	0.6B	0.01	0.09	0.02	0.10	0.02	0.10	0.08	0.03	0.13
	1.7B	0.02	0.08	0.03	0.05	0.01	0.03	0.09	0.04	0.14
	8B	0.04	0.01	0.05	0.01	0.06	0.07	0.15	0.08	0.17
Llama	3.2-3B	0.01	0.06	0.08	0.01	0.07	0.03	0.19	0.04	0.20
Mistral	7B-v0.3	0.00	0.01	0.06	0.04	0.02	0.01	0.16	0.14	0.12
Mean		**0.08**$^{8/8}$	**0.06**$^{8/8}$	**0.10**$^{8/8}$	**0.04**$^{8/8}$	**0.04**$^{8/8}$	**0.04**$^{8/8}$	**0.13**$^{8/8}$	**0.06**$^{8/8}$	**0.16**$^{8/8}$

Table 43: Drifting- and intermediate-feature Cohen's d (gen-1 vs. gen-2+) under three decoding strategies per generator. Drifting features are expected to maintain large d regardless of strategy.

		entropy_mean			TTR			hapax_ratio			entropy_std		
Family	Generator	Nuc.	Gr.	Top-k	Nuc.	Gr.	Top-k	Nuc.	Gr.	Top-k	Nuc.	Gr.	Top-k
Pythia	2.8B	1.20	1.43	0.70	0.77	0.98	0.41	0.67	0.87	0.37	0.39	0.93	0.10
	1B	1.40	0.98	0.83	0.80	0.64	0.41	0.65	0.53	0.34	0.53	0.91	0.13
OLMo	2-1B	1.19	1.29	0.73	1.08	0.89	0.70	1.00	0.80	0.69	0.61	0.95	0.35
Qwen3	0.6B	0.92	0.82	0.80	0.63	0.51	0.58	0.50	0.46	0.52	0.56	0.72	0.30
	1.7B	1.12	1.03	0.99	0.93	0.74	0.85	0.83	0.68	0.80	0.52	0.78	0.32
	8B	0.86	0.98	0.76	0.71	0.76	0.68	0.66	0.76	0.64	0.37	0.71	0.27
Llama	3.2-3B	1.15	1.46	0.77	1.04	1.27	0.73	0.98	1.26	0.71	0.50	1.01	0.21
Mistral	7B-v0.3	0.89	1.31	0.70	0.82	1.04	0.64	0.82	1.03	0.67	0.23	0.62	0.15
Mean		**1.09**$^{8/8}$	**1.16**$^{8/8}$	**0.79**$^{8/8}$	**0.85**$^{8/8}$	**0.85**$^{8/8}$	**0.63**$^{8/8}$	**0.76**$^{8/8}$	**0.80**$^{8/8}$	**0.59**$^{8/8}$	**0.46**$^{8/8}$	**0.83**$^{8/8}$	**0.23**$^{8/8}$

Under greedy decoding (τ=0), shape features remain near-stationary across all eight generators (mean $|d|$=**0.05**, max **0.14**), while core drifting features maintain large effects (mean d=**0.94**), confirming that DDR is a structural property of autoregressive generation independent of sampling truncation. Top-k sampling (k=50) partially attenuates the partition. Shape features show boundary violations in 4/8 generators under top-k; however, these generators exhibit comparable or higher violations under nucleus sampling (Pythia entropy_max: 0.21–0.25 nucleus vs. 0.26–0.27 top-k; OLMo tail_decay: 0.22 under both), indicating model-intrinsic boundary effects rather than strategy-induced degradation. Core drifting effect sizes weaken (mean d=**0.67** vs. **0.90** under nucleus). This modulation is consistent with the mechanistic account in §5: top-k truncation introduces a fixed-width selection window that interacts differently with depth-compressed distributions than nucleus sampling's probability-mass truncation, partially blurring the location/shape boundary. The core DDR partition—drifting features carrying depth signal while shape features provide cross-generator reference—remains intact under both greedy and nucleus sampling and is attenuated but recognizable under top-k. Notably, entropy_std exhibits strong strategy dependence (greedy mean d=0.83; nucleus 0.46; top-k 0.23), suggesting its depth sensitivity is modulated by the token-selection mechanism—deterministic greedy selection amplifies entropy variance across depths, while top-k's fixed window compresses it—further supporting its classification as an intermediate rather than core feature.

aa.5 9-Generator Scale Evaluation

This analysis extends the 8-generator LOGO evaluation to nine generators by adding Gemma-2-2B, a model from a previously unrepresented architecture family (Gemma). All results use class-balanced LOGO; the 8-gen averages here (0.770/0.701) differ slightly from Table 2 (0.762/0.694), which uses an unfiltered protocol. Table 44 reports per-generator LOGO performance under this expanded set.

Table 44: 9-generator LOGO evaluation. Each row holds out the listed generator at test time. 3-class: gen-0 / gen-1 / gen-2+; binary: gen-1 vs. gen-2+. 95% bootstrap CIs in brackets.

Generator	Family	Params	3-class F1	95% CI	Binary F1	95% CI
Pythia-1B	Pythia	1.0B	0.857	[0.848, 0.866]	0.785	—
Qwen3-0.6B	Qwen	0.6B	0.819	[0.807, 0.830]	0.743	—
Pythia-2.8B	Pythia	2.8B	0.812	[0.801, 0.822]	0.728	—
Qwen3-1.7B	Qwen	1.7B	0.811	[0.800, 0.823]	0.746	—
OLMo-2-1B	OLMo	1.5B	0.773	[0.762, 0.784]	0.692	—
Llama-3.2-3B	Llama	3.2B	0.759	[0.748, 0.770]	0.680	—
Qwen3-8B	Qwen	8.2B	0.696	[0.683, 0.709]	0.647	[0.631, 0.664]
Gemma-2-2B	Gemma	2.6B	0.628	[0.622, 0.634]	0.440	[0.432, 0.449]
Mistral-7B	Mistral	7.0B	0.605	[0.593, 0.617]	0.534	—
Average (9-gen)			**0.751**		**0.666**	
Average (8-gen)			**0.770**		**0.701**	

Overall comparison. Adding Gemma-2-2B reduces the mean 3-class LOGO F1 from 0.770 (8-gen) to 0.751 (9-gen), a drop of $\Delta = -0.019$. The per-generator F1 values for the original eight generators shift by at most $[-0.011, +0.002]$, indicating that the inclusion of Gemma data in training does not destabilize existing generalization patterns.

Gemma-2-2B anomaly. Gemma-2-2B is the weakest held-out generator: 3-class F1 = 0.628, binary F1 = 0.440. Per-class breakdown reveals a striking asymmetry: gen-0 F1 = 0.991 and gen-2+ F1 = 0.856, but gen-1 F1 = 0.036; the classifier almost entirely fails to identify first-generation Gemma continuations. This is not a data-quality artifact: the Gemma-2-2B partition contains 4,957 instances with a balanced depth distribution ($\sim$990 per level). A plausible explanation is that Gemma-2's first-generation continuations occupy a region of feature space highly similar to human-written text (gen-0), causing systematic misclassification of gen-1 as gen-0 or gen-2+. This constitutes an architecture-specific generalization limitation rather than a systemic failure of the depth fingerprinting approach. Several architectural differences may explain this behavior: (a) Gemma-2's distinct attention mechanism and SentencePiece tokenizer with a larger vocabulary may produce first-generation continuations whose feature-space signature closely resembles human text, causing systematic gen-0/gen-1 confusion; (b) this anomaly does not affect DDR's validity for the remaining eight generators, where per-generator F1 shifts by at most 0.011 upon Gemma inclusion.

Scale effect. A clear parameter-count gradient emerges across the nine generators: three of four sub-1.7B generators achieve F1 $>$ 0.81 (OLMo-2-1B at 0.773 is the exception), mid-range models (2–3B) cluster at 0.63–0.81, and large models (7B+) fall to 0.60–0.70. Qwen3-8B (8.2B parameters) achieves 3-class F1 = 0.696 under the 9-gen split, compared to 0.701 in the class-balanced 8-gen evaluation; this small difference reflects the changed training composition rather than a property of the Qwen3-8B generator itself.

14B-parameter evidence. Preliminary evidence from a 14.7B generator (Qwen3-14B-Base) confirms that the DDR partition persists at larger scale: Cohen's d for drifting features remains in the large-effect range (d=0.78–1.12), though Spearman $|\rho|$ attenuates by 21–35%, with entropy mean falling just below the drifting threshold ($|\rho|$=0.39 $<$ 0.40, d=0.78). This scale-dependent signal compression is consistent with larger models producing more self-typical outputs that narrow inter-depth contrasts; the partition structure is preserved even as absolute effect sizes compress.

aa.6 Comparison with Concurrent Iterative-Generation Studies

Table 45 compares this work with two concurrent studies on iterative LLM generation. All three examine how repeated inference transforms text, but differ in the chain operator (meaning-preserving vs. open-ended), the analytic lens (factuality decay, convergence dynamics, vs. distributional decomposition), and the downstream goal. Our continuation-chain setting, which imposes no semantic-preservation constraint, permits richer depth signal and enables the location/shape decomposition that defines DDR.

Table 45: Comparison with concurrent iterative-generation studies. All three examine how repeated LLM inference transforms text, but differ in chain operator, analytic lens, and downstream goal.

Study	**Chain Type**	**Feature Space**	**Analysis Goal**	**Generator Scale**	**Steps**	**Core Insight**
Mohamed et al. (2025)	Back-translation (6 bridge langs); rephrasing (ablation)	Surface overlap (BLEU, ROUGE, METEOR, ChrF, BERTScore) + atomic factuality (FActScore)	Factuality degradation rate across iterations	7–9B (3 open models)	100	Factual accuracy decays progressively; rate governed by source–bridge linguistic distance
Geng et al. (2026)	Rephrasing + round-trip translation	String recurrence time, distinct-output count, similarity plateau (METEOR), KL contraction	Convergence & diversity dynamics of inference-time Markov chains	7–8B (3 open + GPT-4o-mini)	50	Greedy decoding collapses to 1–4 recurrent outputs; sampling sustains 10–20 + realizations
Ours	Open-ended continuation (no meaning-preserving constraint)	Token-level NLL statistics (location, scale, shape) + lexical diversity	Depth classification + DDR decomposition into generator & depth effects	0.6–8B (8 models)	5	NLL location drifts monotonically with depth while shape remains invariant—enabling per-generator depth fingerprinting

aa.7 Token-Level NLL Distribution

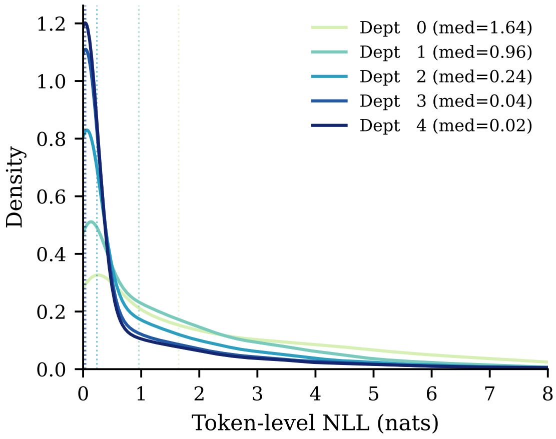

Figure 15: Token-level NLL distribution by generation depth, aggregated across all 8 generators (Pythia-410M reference model). Dashed lines indicate median NLL per depth. The systematic leftward shift demonstrates that deeper-generation text is increasingly predictable.

To visualize how the reference-model surprisal landscape shifts with generation depth, we plot the kernel density estimate of per-token negative log-likelihood (NLL) across depths 0–4 in Figure 15. For each of the 8 base generators, we sample 25 texts per depth (1,000 texts total, $\sim$236k tokens) and compute token-level NLL under the Pythia-410M reference model. The distributions exhibit a pronounced leftward shift as depth increases: gen-0 text (human-written seed) has a median NLL of 1.64 nats, dropping to 0.96 at depth 1 and reaching 0.02 at depth 4, confirming that recursively generated text becomes increasingly predictable to the reference model. The monotonic compression of the NLL distribution provides a mechanistic explanation for the entropy-based features that drive our depth classifier.